ELSEVIER

Physica D 127 (1999) 146–176

PHYSICA D

Large-amplitude solutions to the Sivashinsky and Riley–Davis equations for directional solidification

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Received 8 July 1996; received in revised form 30 May 1997; accepted 27 October 1998
Communicated by C.K.R.T. Jones

Abstract

Directional solidification of a dilute binary alloy is often characterized by the appearance of deep-root, finger-type interfaces. As a model of this phenomenon, we investigate the large-amplitude behavior of long-wave evolution equations previously derived for directional solidification. In this paper we conduct a detailed asymptotic and numerical study of the periodic solutions of the Sivashinsky and Riley–Davis equations. The Sivashinsky equation, which describes the limit of small segregation coefficient, is shown to form deep-root solutions which can be described in terms of elliptic functions. However, this equation is known to be ill-posed in the sense that the solution branches are unstable, and large classes of initial conditions lead to finite-time blow-up. By considering asymptotically large surface energy, Riley and Davis derived an evolution equation which may be considered a regularizing perturbation of the Sivashinsky equation. This equation has steady, large-amplitude solutions which are linearly stable. In the absence of solutal segregation, we may describe these solutions asymptotically; in this limit, amplitudes and wavelengths may become arbitrarily large. A stability analysis suggests that with periodic boundary conditions the solution may coarsen until a single finger fills the domain. If segregation effects are included, both the amplitude and wavelength are bounded independent of domain size. This wavelength bound gives the system a finger-width selection mechanism associated with a tip-splitting bifurcation that has also been observed in experiments. ©1999 Elsevier Science B.V. All rights reserved.

Keywords: Sivashinsky equation; Riley–Davis equation; Directional solidification

1. Introduction

The complex dynamics exhibited by a solid–liquid interface during the directional solidification of a binary alloy has been studied for more than 30 years. As is the case with most physical processes, one must typically make

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PII: S0167-2789(98)00281-4

numerous approximations and idealizations in order to formulate a manageable mathematical model. In order to extract a description of the behavior of the interface, one must then make further simplifications or restrict oneself to certain limiting cases. As such, a full understanding of such phenomena as the formation of deep-rooted cells or the growth of dendrites remains an elusive goal.

Most of the early analytical work on this subject focused on the identification of sufficient conditions for a morphological breakdown of a planar interface in the absence of convection, and a description of the resulting cellular structures. The original model proposed by Mullins and Sekerka [17] in their 1964 paper consisted of diffusion equations for solute concentration and temperature in both the solid and liquid phases with appropriate far-field conditions, and interfacial conditions corresponding to the assumption of local thermodynamic equilibrium and conservation of energy and solute across the interface. They showed that for this model the basic planar state may be linearly unstable to spatially periodic disturbances.

The bifurcation problem, which attempts to describe the periodic structure of the slightly deformed interface, was first analyzed for the Mullins and Sekerka model in two dimensions by Wollkind and Segel [36], and extended to three dimensions by Sriranganathan et al. [33]. The Mullins and Sekerka analysis has been extended to incorporate additional features such as nonequilibrium effects [8], solute diffusion in the solid [5,18,34], and wall effects [35]. For a more thorough treatment of these models, the reader is referred to an excellent review article by Coriell and McFadden [7] on the morphological stability problem.

The purpose of this paper is to provide greater mathematical and physical insight into the appearance of large-amplitude interfaces whose description lies beyond the scope of linear stability theory or weakly nonlinear analysis. We focus our attention on a further analysis of the previous work in long-wave theory for this problem. The essence of long-wave theory lies in the identification of a physical parameter limit for which the instability wavelength becomes asymptotically large. In this limit, it may be possible to obtain an equation which represents a leading-order description of the motion of the interface, decoupled from the other variables in the problem. This *long-wave evolution equation* represents a significant simplification of the full model and is much more amenable to analytical and numerical study.

The first use of this theory in the context of directional solidification was conducted by Sivashinsky [31] in 1983, who identified asymptotically small segregation coefficient as a limit leading to long-wave instabilities. The resulting evolution equation was found to describe a *subcritical* bifurcation from the planar state to an unstable periodic structure. Subsequent studies by Novick-Cohen demonstrated that all periodic solutions to the Sivashinsky equation are unstable [20] and that many initial conditions lead to finite-time blow-up [21,22] (see also the work of Bernoff and Bertozzi [2]).

In a later study, Brattkus and Davis [3] investigated the absolute stability limit and derived a long-wave equation which described two-dimensional structures that bifurcate *supercritically* from the planar state.

Subsequently, Riley and Davis [29] identified yet a third distinguished limit which led to long-wave disturbances, that of small segregation coefficient and large surface energy. The resulting evolution equation was found to describe a transition between subcritical and supercritical bifurcations, and recovered the Sivashinsky equation in the appropriate limit.

Analysis of directional solidification has also been extended by Novick-Cohen and Sivashinsky [26] to include latent heat effects, by Young et al. [39] to include anisotropic attachment kinetics, and by Young and Davis [38] to include end effects. Novick-Cohen [19] studied the limit of weak thermal gradient and rapid transport in a model which included the effect of attachment kinetics, and demonstrated that the motion of the interface was described by a Kuramoto–Sivashinsky equation. Recent studies have also considered the bifurcation of periodic solutions in the Sivashinsky, Riley–Davis and Brattkus–Davis equations [10,32]. These studies generally concentrated on bifurcation theory, and the bounded dynamics associated with the various amplitude equations.

In a recent study, Metzener [14] re-considered the Brattkus–Davis absolute stability limit and generalized their equation to include the effects of attachment kinetics and a weak flow. He then proceeds to describe large-amplitude solutions of this equation, taking into account the nonlinear thermodynamic effects which may be experienced for large amplitude fluctuations. In particular, he obtains deep-groove solutions analytically.

In this paper we conduct a thorough investigation of the Sivashinsky and Riley–Davis equations. In particular, we analytically and numerically compute large-amplitude time-independent solutions to these equations and investigate their stability. For the Sivashinsky equation it is known that all periodic solutions are unstable [20]. In contrast, for the Riley–Davis equation, we use asymptotic matching to describe deep-root solutions which may be stable. These results suggest that the additional terms in the Riley–Davis equation may regularize the dynamics of the Sivashinsky equation.

In the context of the solidification problem, the analysis contained here remains a valid description of the dynamics of the underlying model provided that large spatial gradients remain small compared to the instability wavelength. Thus, this work may provide valuable insight into the development of deep-root interfaces in experimental settings and in numerical simulations of the full models.

In Section 2, we introduce the Sivashinsky and Riley–Davis equations and the relevant physical parameters. We then identify an important zero-mean condition and outline our method for studying the equations. In Section 3, we treat the Sivashinsky equation in detail, including a summary of the results on periodic solutions and their stability, and the blow-up results previously obtained. In Section 4, we investigate the Riley–Davis equation, including a description of deep-root interfaces and a detailed linear stability analysis. In Section 5, we summarize our results. Finally, Appendix A gives the details of the numerical methods employed here.

2. Formulation/zero mean condition

Fig. 1 shows a typical configuration for a directional solidification experiment which may be modeled as two-dimensional. Two closely spaced parallel plates enclose a binary mixture which is then drawn at a fixed pulling

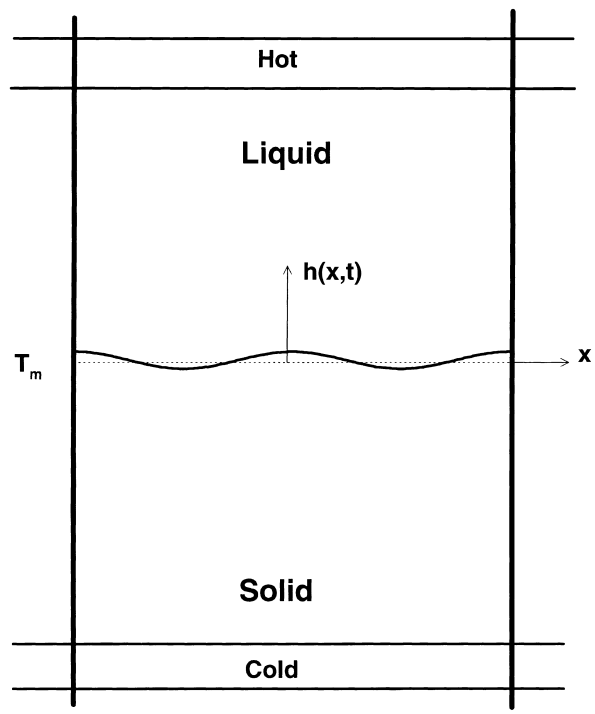

Fig. 1. Directional solidification. In a typical experimental setting, a binary alloy occupies the region between two parallel plates. The initial liquid material is pulled at fixed pulling speed V through a temperature gradient which solidifies the material. Long-wave theory may be used to predict the deformation of the solid–liquid interface.

speed V through a temperature gradient. The stability of the planar interface, and the structure and evolution of any subsequent deformations, are assumed to be governed solely by the pulling speed, the magnitude of the temperature gradient, the solute concentration, and the relevant material parameters, while the effects of other external factors such as end walls are ignored. The model adopted by both Sivashinsky and Riley and Davis to describe this process is the one originally proposed by Mullins and Sekerka [17] and later modified by Wollkind and Segel [36] and Langer [13]. Solute diffusion in the solid is neglected, while the "frozen temperature" approximation proposed by Langer results in a decoupling of the temperature field from the governing equations. The reader is directed to the above references for a complete formulation of this model, which will not be presented here. The scaled form of this model as analyzed by Riley and Davis [29] contains three nondimensional parameters: a segregation coefficient k, which measures the ratio of the solute concentration at the interface on the solid and liquid sides; a morphological parameter M, and a scaled surface energy Γ. The case $k = 0$ corresponds to complete solute rejection by the solidifying material (the "segregation-free" case), while $k = 1$ corresponds to complete "solute trapping".

Sivashinsky [31] identifies the limit of small segregation coefficient ($k \ll 1$) and near-unity morphological parameter ($1 - M^{-1} \ll 1$) as a distinguished limit which leads to long-wave instabilities, and finds that the motion of the interface in this case is governed to leading order by the evolution equation

$$H_T + \Gamma H_{XXXX} + H_{XX} + \frac{k}{[\mu(M)]^2} H = \left(\frac{H^2}{2}\right)_{XX}, \tag{2.1}$$

where $\mu(M) \equiv (1 - M^{-1})$, the quantities Γ and $k[\mu(M)]^{-2}$ are O(1), and X, T, and H are rescaled temporal, spatial, and amplitude deformation variables defined, respectively, in terms of the slowly varying original variables $x = [\mu(M)]^{-1/2}X$, $t = [\mu(M)]^{-2}T$, and the small-amplitude deformations given by $h(x,t) = [\mu(M)]H(X,T)$.

Alternatively, Riley and Davis [29] identified the limit of small segregation coefficient ($k \ll 1$) and large surface energy ($\Gamma \gg 1$) as an alternative long-wave limit, and found that to leading order

$$H_T - MH_{TXX} + MH_{XXXX} + \delta H_{XX} + \Gamma kH = \left(\frac{H^2}{2}\right)_{XX} - M\left(\frac{H_X^2}{2}\right)_{XX}, \tag{2.2}$$

where $\delta \equiv M - 1 - \Gamma kM$, the quantities H, M, and Γk are O(1), and $x = \Gamma^{1/2}X$, $t = \Gamma T$, $h(x,t) = H(X,T)$.

Through a suitable rescaling of (2.2), we may express the Riley–Davis equation in a form more convenient for our analysis. We let

$$x = \left(\frac{\delta}{M}\right)^{1/2} X, \qquad t = \frac{\delta^2}{M} T, \qquad h = \delta H.$$

This yields

$$h_t - \delta h_{txx} + h_{xxxx} + h_{xx} + \kappa h = \left(\frac{h^2}{2}\right)_{xx} - \delta\left(\frac{h_x^2}{2}\right)_{xx}, \tag{2.3}$$

where κ is a rescaled segregation coefficient defined by $\kappa \equiv \Gamma kM/\delta^2$. When $\delta = 0$, we obtain

$$h_t + h_{xxxx} + h_{xx} + \kappa h = \left(\frac{h^2}{2}\right)_{xx}, \tag{2.4}$$

which may be seen as a rescaled version of the Sivashinsky equation (2.1). Consequently, the Sivashinsky equation may formally be considered as the limit of the Riley–Davis Equation as $\delta \to 0$; we shall argue that these apparently small terms tend to have a regularizing effect on the observed dynamics. Herein we shall consider the Riley–Davis and Sivashinsky equations only in the forms (2.3) and (2.4), respectively.

By considering a spatially periodic solution with period $2L$ and integrating over a period, we obtain the following solvability condition for both the Sivashinsky and Riley–Davis equations:

$$\bar{h}_t = -\kappa \bar{h}, \tag{2.5}$$

where $\bar{h}(t) \equiv (1/2L)\int_0^{2L} h \, \mathrm{d}x$ is the mean of h over the spatial period. Thus,

$$\bar{h}(t) = \bar{h}(0)\, \mathrm{e}^{-\kappa t}. \tag{2.6}$$

From (2.6) we see that for $\kappa \neq 0$, any steady, periodic solution of (2.3) or (2.4) must have zero mean, analogous to the solid–liquid interface remaining near the location of the equilibrium melting temperature in the externally imposed temperature gradient.

Substantial progress in studying solutions of the Sivashinsky and Riley–Davis equations may be made by first ignoring segregation effects ($\kappa = 0$), corresponding to complete solute rejection. In this limit, the time-independent versions of (2.4) and (2.3) are each thrice integrable, making an analytical and numerical study much more tractable. Unfortunately, the zero-mean condition from Eq. (2.6) is lost; in fact, any solution with constant h now satisfies Eq. (2.3).[4] For this case, each equation has a two-parameter family of steady, periodic solutions; e.g. for ranges of interval length $2L$ and mean value $\bar{h}$, one may find periodic solutions (cf. [24,25]). By contrast, for $\kappa \neq 0$, each equation has a one-parameter family of solutions, all of which have zero mean. As $\kappa \to 0$, a branch of these solutions approaches the one-parameter subset of the $\kappa = 0$ solutions which have zero mean. This result was demonstrated by Novick-Cohen and Grinfeld [23] for the Sivashinsky equation (2.4) by integrating the equation over a spatial period and invoking the implicit function theorem; their argument is equally valid when applied to the Riley–Davis equation (2.3).

In Section 3 and 4, we treat in detail the Sivashinsky and Riley–Davis equations, respectively. For each equation, we shall first consider zero-mean periodic solutions for the $\kappa \to 0$ case, and then compare these results to the full equations by considering nonzero κ as a small perturbation for large-amplitude solutions. In the case of the Sivashinsky equation, such an analysis has already been undertaken by Novick-Cohen and her collaborators [20,23–25]; we review their results and extend them slightly. In Section 4, we will see that the Riley–Davis equation may be considered a regularization of the instability and blow-up previously described for the Sivashinsky equation. Large-amplitude, segregation-free solutions are described to leading order using asymptotic matching. Results for $\kappa > 0$ suggest a coarsening process which may lead to stable, deep-root solutions. The stability of these solutions to disturbances with the same or twice the period is examined numerically.

3. Sivashinsky equation

Periodic solutions to the Sivashinsky equation have been studied previously by Novick-Cohen and her collaborators [20,23–25]. In particular, they describe steady solutions in terms of elliptic integrals in the limit $\kappa \to 0$ [24,25]. They proved the existence of a unique simple zero-mean solution for every period less than 2π. They also showed that these solutions persist for sufficiently small κ [23]. Novick-Cohen [20] used variational methods to prove that all periodic solutions are unstable for any $\kappa \geq 0$.

In fact, the Sivashinsky equation manifests finite-time blow-up. This was first shown by Novick-Cohen [21], and these results were sharpened by Bernoff and Bertozzi [2] using the Fourier method of Palais [27]. Basically, it was shown that Eq. (2.4) with periodic boundary conditions manifests finite-time blow-up for (i) a class of initial conditions with arbitrarily small amplitude if the zero state is linearly unstable and (ii) a class of initial conditions

[4] This arbitrary location of the interface reflects a one-parameter family of solutions in the one-sided model originally considered by Riley & Davis with $\kappa = 0$. However, as κ is used in the rescaling of the original solidification equations, this represents a singular limit of the full problem.

of finite amplitude if the zero state is linearly stable. Numerical results suggest that generically this blow-up is pointwise and self-similar; a wealth of detail may be found in the paper by Bernoff and Bertozzi [2].

Below we first discuss the bifurcation of periodic solutions from the zero solution. In the case $\kappa \to 0$, a complete description of the steady solutions in terms of elliptic functions is possible [24,25]. For $\kappa > 0$, we argue that at large amplitudes, there is a branch of solutions that is described to leading order by the $\kappa \to 0$ solutions. Our numerical results support these conclusions. Although these solutions are all unstable, they provide a framework for analyzing the periodic solutions of the Riley–Davis equation.

3.1. Linear theory and bifurcation analysis

A linear stability analysis of the Sivashinsky equation (2.4) yields the result

$$\sigma = -q^4 + q^2 - \kappa, \tag{3.1}$$

where σ is the growth rate of a sinusoidal perturbation of wave number q. For each $0 < \kappa < 1/4$, there exists a finite band of wave numbers for which the planar front is unstable, and two positive real roots of $\sigma = 0$, indicating primary bifurcation points q_+ and q_-:

$$(q_\pm)^2 = \frac{1 \pm \sqrt{1 - 4\kappa}}{2}. \tag{3.2}$$

The planar front is unstable when $q_- < q < q_+$, and is stable otherwise. Note that when $\kappa = 0$ there is only one critical wave number q_+. Weakly nonlinear analysis [10,12,20,29,32] reveals that the bifurcation at q_+ is subcritical to a branch of unstable, periodic solutions which exist for $q > q_+$, while the bifurcation at q_- is supercritical when $0 < \kappa < 4/25$ and subcritical when $4/25 < \kappa < 1/4$. In the case when the bifurcation at q_- is supercritical, the solution is initially unstable to a perturbation of half the period. Consequently, all the solutions branches are initially unstable, in agreement with the results of Novick-Cohen [20].

In the remainder of Section 3, we will consider steady, large-amplitude solutions to the Sivashinsky equation which may be found along the continuation of the q_+ branch. We first review the limit $\kappa \to 0$, for which these solutions have already been described analytically by Novick-Cohen and Peletier [24,25]. The full equation for $\kappa > 0$ is then investigated, for which we extend previous results to include all values of κ between 0 and 1/4.

3.2. Results for $\kappa \to 0$

Steady solutions to the Sivashinsky equation in the limit of zero segregation coefficient ($\kappa = 0$) have been previously described by Novick-Cohen and Peletier [24,25]. Below we summarize their results in terms of our scalings. The elliptic function description of the solutions will then be used in Section 3.3 along with perturbation theory to derive asymptotic results for $\kappa > 0$.

We seek spatially periodic, time-independent solutions to

$$h_{xxxx} + h_{xx} = \left(\frac{h^2}{2}\right)_{xx}, \tag{3.3}$$

with the requirement

$$\frac{1}{2L}\int_0^{2L} h \, \mathrm{d}x = 0. \tag{3.4}$$

The one-parameter family of solutions which bifurcates from $q_+ = 1$ was calculated numerically using a shooting method; Fig. 2(a) shows this branch in amplitude–wave number space. Several solution profiles at representative

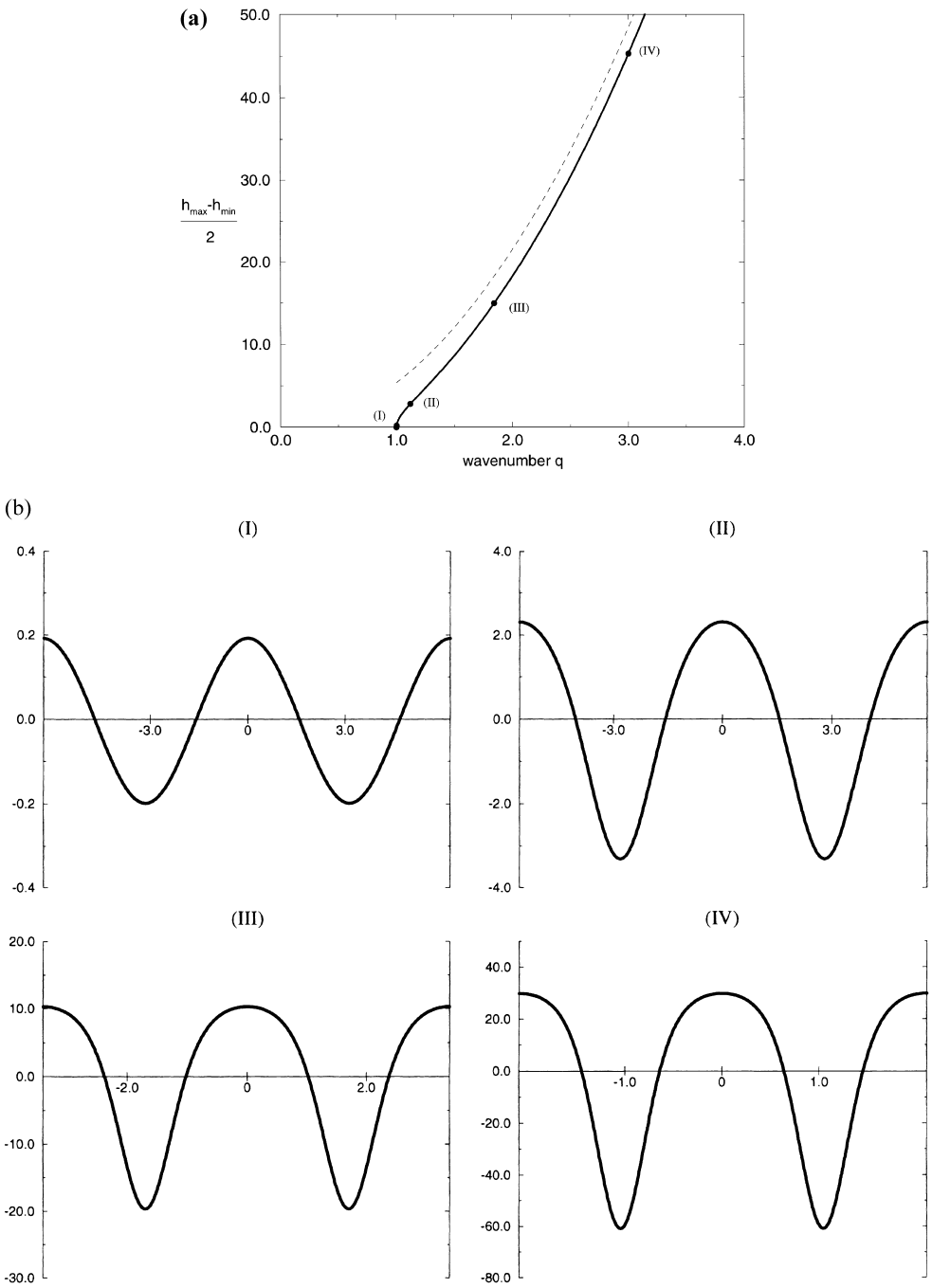

Fig. 2. Sivashinsky equation for $\kappa = 0$. (a) The primary branch of steady, periodic solutions which bifurcates subcritically from the planar solution. For large amplitude, the branch approaches the parabolic profile (dashed line) given by Eq. (3.15). (b) Representative solution profiles along the branch; the sinusoidal structure near the bifurcation point (I) gives way to a finger-like structure at larger amplitudes (II–IV); these solutions are described analytically by Eq. (3.11).

points along the branch as indicated are shown in Fig. 2(b). The large-amplitude solutions exhibit the deep-grooved, finger-type solutions seen in past numerical simulations of the Sivashinsky equation [2,9,30,37].

It was shown by Novick-Cohen and Peletier [25] that (3.3) may be integrated three times and reduced to the potential problem

$$(u_x)^2 + f(u) = d, \qquad f(u) \equiv -\frac{u^3}{3} + cu, \tag{3.5}$$

where $u \equiv h - 1$, and c and d are constants of integration. Through a suitable rescaling, one parameter can be eliminated:

$$(U_X)^2 + F(U) = D, \qquad F(U) \equiv -\frac{U^3}{3} + U, \tag{3.6}$$

where

$$U \equiv c^{-1/2}u, \quad X \equiv c^{1/4}x, \quad D \equiv c^{-3/2}d, \quad -\tfrac{2}{3} \le D \le \tfrac{2}{3}. \tag{3.7}$$

Novick-Cohen and her collaborators [23–25] investigated the above system and demonstrated the existence of a unique simple zero-mean solution for any period less than 2π (by simple we mean here that the solution is not the repetition of multiple periods). In general, periodic solutions to (3.6) are restricted to $-2/3 \le D \le 2/3$, as determined by the local extrema of the potential $F(U)$, though Novick-Cohen et al. actually showed that zero-mean solutions exist only for $-2/3 \le D \le D^*$, where D^* is slightly less than 2/3. If one expresses the quantity $F(U) - D$ in factored form,

$$F(U) - D = -\frac{1}{3}(U-\alpha)(U-\beta)(U-\gamma), \quad \alpha \ge \beta \ge \gamma, \tag{3.8}$$

their elliptic function solution of (3.6) may be conveniently expressed as follows:

$$U = \gamma + (\beta - \gamma)\,\mathrm{sn}^2\left(\frac{\sqrt{\alpha-\gamma}}{2\sqrt{3}}X, \eta\right), \tag{3.9}$$

where $\mathrm{sn}(\xi, \eta)$ is a Jacobian elliptic function and the parameter η, known as the *modulus*, is defined by

$$\eta = \sqrt{\frac{\beta-\gamma}{\alpha-\gamma}}, \quad 0 \le \eta \le 1. \tag{3.10}$$

In terms of the unscaled variables, Eq. (3.9) is equivalent to

$$h = 1 + c^{1/2}\left[\gamma + (\beta-\gamma)\,\mathrm{sn}^2\left(\frac{\sqrt{\alpha-\gamma}}{2\sqrt{3}}c^{1/4}x, \eta\right)\right]. \tag{3.11}$$

For a given choice of D, (3.8) determines α, β, and γ, and therefore η. The desired one-parameter family of solutions is found by varying the parameter D, then choosing the constant c to satisfy the zero-mean condition (3.4). One finds that

$$c = \frac{[K(\eta)]^2}{[\alpha K(\eta) - (\alpha-\gamma)E(\eta)]^2}, \tag{3.12}$$

where $K(\eta)$ and $E(\eta)$ are the complete elliptic integrals of the first and second kind, respectively.

A thorough discussion of the properties of elliptic integrals and their asymptotic behaviors may be found in [4]. The periodic function $\mathrm{sn}(\xi, \eta)$ satisfies $-1 \le \mathrm{sn}(\xi, \eta) \le 1$, and $\mathrm{sn}^2(\xi, \eta)$ passes through one complete cycle as the spatial variable ξ varies by an amount $2K(\eta)$. Thus, denoting the period of h as $2L$, it follows from Eq. (3.11) that

$$L = \frac{2\sqrt{3}K(\eta)}{c^{1/4}\sqrt{\alpha-\gamma}} \tag{3.13}$$

and

$$h_{\max} - h_{\min} = c^{1/2}(\beta - \gamma). \tag{3.14}$$

In the limit $D + 2/3 \ll 1$, it follows from (3.8) and (3.10) that $\eta \ll 1$. In this limit, $\mathrm{sn}(\xi, \eta) = \sin(\xi) + \mathrm{O}(\eta^2)$, for which Eq. (3.11) describes the subcritical pitchfork bifurcation predicted by weakly nonlinear analysis of the Sivashinsky equation.

Also shown by Novick-Cohen and Peletier [25] is that the branch of periodic, zero-mean solutions does not range over the entire interval $0 \le \eta \le 1$. For $\eta = \eta^* \approx 0.9804$, corresponding to $D = D^* \approx 0.6631$, the denominator in (3.12) equals zero, and the rescalings in (3.7) break down. Asymptotically large-amplitude solutions are described by letting $\eta \to \eta^*$ from below. In this limit c approaches infinity, and as the quantities $K(\eta)$, $(\alpha - \gamma)$, and $(\beta - \gamma)$ all remain bounded and nonzero, it follows from (3.13) and (3.14) that asymptotically large amplitudes scale as the square of the wave number. Specifically, by plugging $D = D^*$ into (3.8) and computing α, β, and γ, and tabulating $K(\eta^*)$ and $E(\eta^*)$, it is seen that

$$h_{\max} - h_{\min} \approx 10.7306 q^2, \tag{3.15}$$

where $q \equiv \pi/L$ is the wave number. (3.15) is demonstrated numerically in Fig. 2(a). This result represents an extension of the analysis of Novick-Cohen and Peletier, who precisely described the asymptotic relationship between the wave number and the *mean* of h. However, (3.15) is readily attainable from their analysis.

3.3. *Results for* $0 < \kappa < 1/4$

In Fig. 3, branches of steady solutions for the full Sivashinsky equation are traced numerically for various nonzero values of κ, and are shown along with the zero-mean branch for $\kappa = 0$. As mentioned previously, for each $\kappa \neq 0$, there is an additional branch of solutions corresponding to a second primary bifurcation point q_- given by (3.2). In the figure, the q_- branches appear to the left of $q = \sqrt{2}/2$. Our numerical results indicate that the q_- branches typically terminate at finite amplitude, with the terminal point corresponding to a secondary, period n-tupling bifurcation along the q_+ branch, where n grows larger as κ becomes small. As the q_- branches do not describe solutions of asymptotically large amplitude, these branches are not of primary interest here.

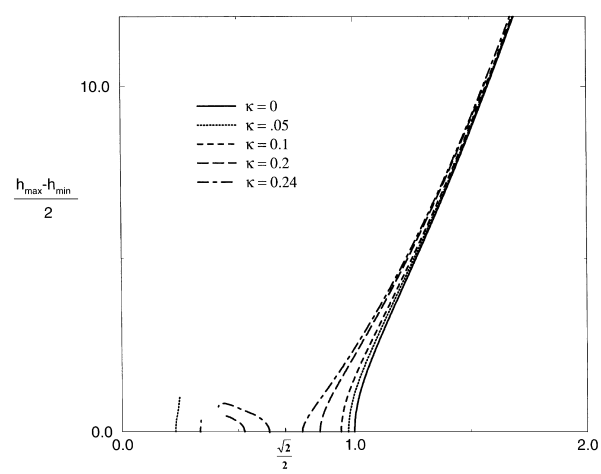

Fig. 3. Sivashinsky equation, $0 \le \kappa \le 1/4$. For each $\kappa \neq 0$, there are two primary bifurcation points. The q_- branches (to the left of $q = \sqrt{2}/2$ in the figure) terminate at finite amplitude, while the convergence of the q_+ branches to the $\kappa = 0$ branch indicates that the $\kappa \neq 0$ perturbation is always small for large amplitudes.

In accordance with linear theory, these results clearly indicate that the κh term in the Sivashinsky equation may be co-dominant for sufficiently small-amplitude solutions. However, the q_+ branches all appear to converge to the $\kappa = 0$ branch, suggesting that for large amplitudes, the κh term is subdominant for *any* value of κ. This suggests an extension of the previous result which says that the branch of solutions persist for sufficiently small κ [23].

To check this result, we substitute the segregation-free solution (3.11) into the full equation (2.4) and recall that large-amplitude solutions correspond to $c \gg 1$ in (3.11). Using the boundedness of α, β, γ, and $\mathrm{sn}(v, k)$ and its derivatives in this limit, and denoting the segregation-free solution as $\bar{h}$, we see that

$$\bar{h}_{xxxx} \sim c^{3/2}, \qquad \bar{h}_{xx} \sim c, \qquad \kappa\bar{h} \sim \kappa c^{1/2}, \qquad \left(\frac{\bar{h}^2}{2}\right)_{xx} \sim c^{3/2}. \tag{3.16}$$

A comparison of the sizes of the various terms in (3.16) suggests that the κh term is subdominant for large-amplitude solutions. By letting

$$h = \bar{h} + \kappa h_1 + \cdots \tag{3.17}$$

and substituting into (2.4), we can approximate the size of the leading-order correction h_1 as a function of c. The $\mathrm{O}(\kappa)$ equation is

$$h_{1_{xxxx}} + h_{1_{xx}} = (\bar{h}h_1)_{xx} - \bar{h}, \tag{3.18}$$

where $\bar{h}$ is given by Eq. (3.11). Substituting (3.11) into (3.18), a further dimensional analysis suggests $h_1 \sim H(c^{1/4}x)$, where $H \sim 1$, consistent with our numerical observation that the vertical distance between the branches in Fig. 3 scales with κ at large amplitude.

3.4. Conclusions

In summary, we have examined the periodic solutions of the Sivashinsky equation. In the limit of large amplitude, there exists a branch of solutions with decreasing period that appear to persist for all κ. However, it is known that all the periodic solutions are unstable and that large classes of initial conditions manifest at finite-time blow-up.

In the next section, we will present evidence suggesting that the Riley–Davis equation acts as a regularization of the Sivashinsky equation.

4. Riley–Davis equation

The Riley–Davis equation (2.3),

$$h_t - \delta h_{txx} + h_{xxxx} + h_{xx} + \kappa h = \left(\frac{h^2}{2}\right)_{xx} - \delta\left(\frac{h_x^2}{2}\right)_{xx}$$

is distinguished from the Sivashinsky equation primarily in its ability to describe *stable* time-independent solutions. Note that when $\delta \to 0$ the Sivashinsky equation (2.4) is recovered, while $\delta > 0$ introduces an additional nonlinearity proportional to $(h_x^2)_{xx}$ which has a regularizing effect on the periodic solutions to the equation. This regularization is manifested in two ways. First, as shown by Riley and Davis [29], the initial bifurcation from the planar front at q_+ may be super- or subcritical. Second, as we shall see, when the bifurcation is subcritical, the initially unstable branch of solutions eventually undergoes a stabilizing saddle-node bifurcation.

Our analysis in the remainder of this section will also reveal the following results:

(i) For any finite amplitude, as $\delta \to 0$ the periodic solutions of the Riley–Davis equation approach those of the Sivashinsky equation. However, at larger amplitudes these branches always undergo a stabilizing saddle-node bifurcation.

(ii) When segregation effects are ignored ($\kappa \to 0$), amplitudes and wavelengths may become arbitrarily large. These large-amplitude, deep-root solutions may be described asymptotically by analyzing a potential problem, as they were for the Sivashinsky equation.
(iii) When segregation effects are ignored ($\kappa \to 0$), large amplitude solutions are always unstable to a solution with twice the spatial period. This suggests a *coarsening* process, by which fingers will join until a single finger dominates the domain.
(iv) Inclusion of segregation effects ($\kappa > 0$) leads to bounds on both the amplitude and wavelength, and a period-halving bifurcation which may be associated with tip-splitting or tip-joining. A range of stable solutions are observed, suggesting that segregation effects determine the characteristic width of the observed fingers.

In particular, the open problems of coarsening (in the absence of segregation effects) and global existence and uniform bounds on the temporal evolution (in the presence of segregation effects) should provide fertile grounds for future analysis.

4.1. Bifurcation analysis

We begin with a modified version of the bifurcation analysis done by Riley and Davis in their original paper [29]. As with the Sivashinsky equation, there exists for $\kappa < 1/4$, a finite band of wave numbers for which the planar front is unstable. The width of this band and the location of the primary bifurcation points q_+ and q_- are independent of δ and are given by (3.2), identical to that for the Sivashinsky equation (2.4), while the growth rate is modified slightly:

$$\sigma = \frac{-q^4 + q^2 - \kappa}{1 + \delta q^2}. \tag{4.1}$$

One may proceed with a weakly nonlinear analysis of the bifurcation by considering small-amplitude deformations of the form

$$h = \varepsilon A(\tau)\cos qx + \varepsilon^2 B(\tau)\cos 2qx + \cdots, \quad \tau \equiv \varepsilon^2 t, \quad q \equiv q_* + \varepsilon^2 Q, \quad \varepsilon \ll 1, \tag{4.2}$$

where $q_* = q_+$ or q_-. At $O(\varepsilon^3)$, a solvability condition requires

$$(1 + \delta q_*^2)A_\tau = 2q_*(1 - 2q_*^2)QA - \left[\frac{q_*^4\left(\delta q_*^2 - 1/2\right)(\delta q_*^2 + 1)}{12q_*^2 - 15\kappa}\right]A^3. \tag{4.3}$$

The transition from a subcritical to a supercritical bifurcation occurs when the sign of the Landau coefficient changes, and in this case we find that the bifurcation at q_+ is supercritical when

$$\delta > \frac{1}{1 + \sqrt{1 - 4\kappa}}, \tag{4.4}$$

and subcritical otherwise, while the bifurcation at q_- is supercritical if

$$0 < \kappa < \frac{4}{25} \quad \text{and} \quad \delta < \frac{1}{1 - \sqrt{1 - 4\kappa}}, \tag{4.5}$$

or if

$$\frac{4}{25} < \kappa < \frac{1}{4} \quad \text{and} \quad \delta > \frac{1}{1 - \sqrt{1 - 4\kappa}}, \tag{4.6}$$

and subcritical otherwise.

These conditions are illustrated in Fig. 4. For the bifurcation at q_- (Fig. 4(a)), the $\kappa - \delta$ parameter space is divided into four regions according to (4.5) and (4.6). At q_+ (Fig. 4(b)), we see that the bifurcation is *always* supercritical

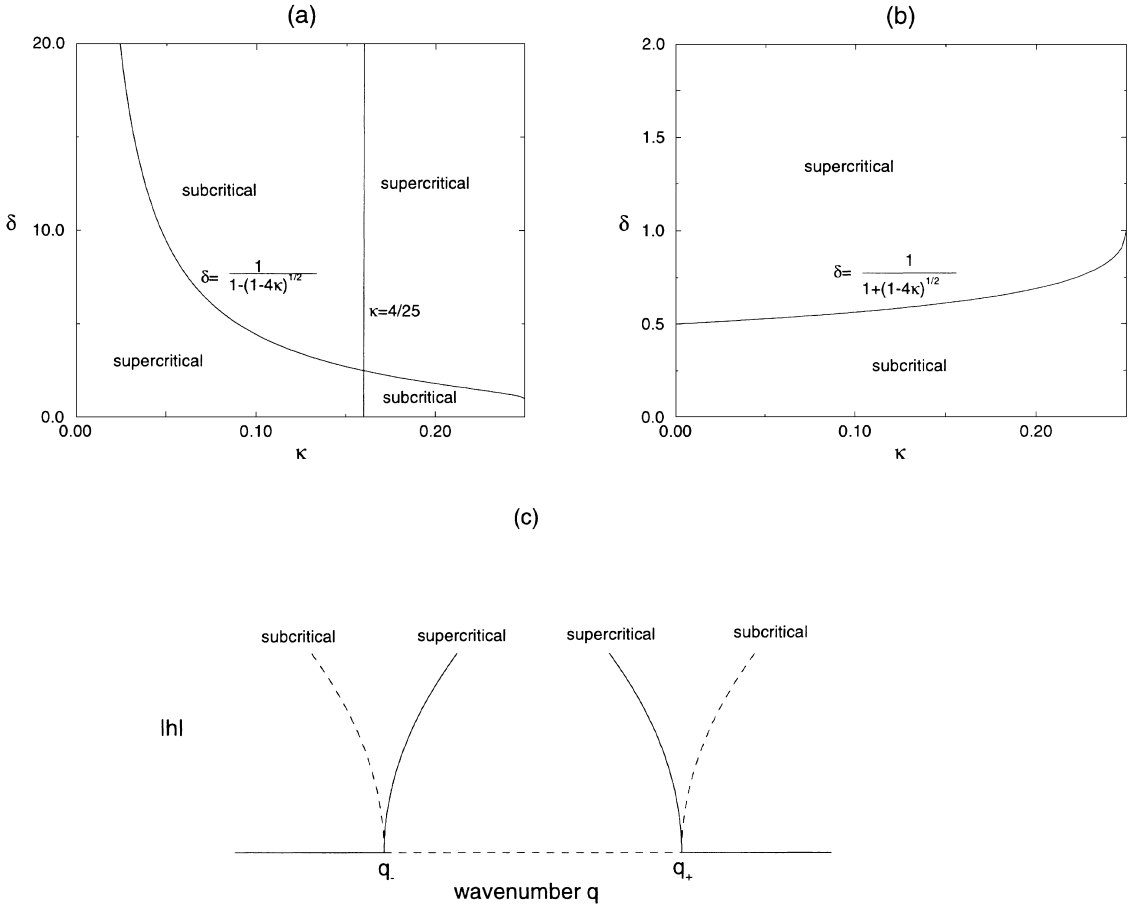

Fig. 4. (a) Unfolding diagram for the Riley–Davis equation (2.3) at q_-; (b) unfolding diagram at q_+; (c) bifurcation structure near the primary points: the planar front is unstable between q_- and q_+.

when $\delta > 1$ and *always* subcritical when $\delta < 1/2$, while for $1/2 < \delta < 1$, the nature of the bifurcation depends on the relationship between κ and δ. Also, the corresponding results for the Sivashinsky equation may be inferred here by looking along the horizontal axis ($\delta = 0$) in Fig. 4(a) and (b). The bifurcation at q_+ is always subcritical, while the bifurcation at q_- is supercritical when $\kappa < 4/25$.

By analogy with the bifurcation analysis of Knobloch [12] in long-wavelength convection, one may also show that even when the bifurcation at q_- is supercritical, it may be unstable to a perturbation of twice the wave number. This turns out to be the case for the upper right-hand region of Fig. 4(a), for which $\kappa > 4/25$ and $\delta > 1/(1-(1-4\kappa)^{1/2})$. Thus, only a small class of the q_- branches describe solutions which may be stable.

4.2. Results for $\kappa \to 0$

Similar to our analysis of the Sivashinsky equation, we first analyze the Riley–Davis equation (2.3) by ignoring segregation effects entirely ($\kappa = 0$) and seeking time-independent solutions to

$$h_{xxxx} + h_{xx} = \left(\frac{h^2}{2}\right)_{xx} - \delta\left(\frac{h_x^2}{2}\right)_{xx} \tag{4.7}$$

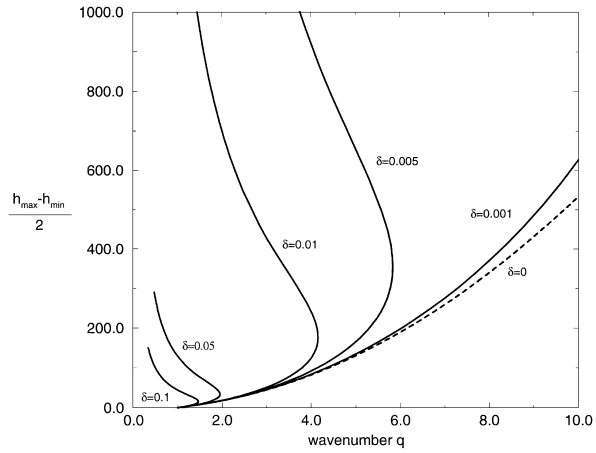

Fig. 5. Riley–Davis equation for $\kappa = 0$. The segregation-free Riley–Davis equation may be viewed as a singular perturbation of the Sivashinsky equation (dotted line). The Riley–Davis branches experience a stabilizing saddle-node bifurcation at an amplitude that scales as $1/\delta$; past the saddle node, amplitudes and wavelengths become arbitrarily large. The branches corresponding to $\delta = 0.1$ and $\delta = 0.05$ do not terminate as indicated in the figure, but rather could not be traced any further due to numerical stiffness; the saddle node for the $\delta = 0.001$ branch occurs outside the bounds of the figure.

with the requirement

$$\frac{1}{2L}\int_0^{2L} h \, \mathrm{d}x = 0. \tag{4.8}$$

Solution branches to (4.7), (4.8) were computed numerically for various nonzero values of δ and are shown in Fig. 5, along with the $\delta = 0$ branch, which corresponds to the segregation-free Sivashinsky equation (3.3). The regularizing effect of the additional nonlinearity is clearly seen in the figure; the initially subcritical branch of solutions undergoes a stabilizing saddle-node bifurcation. The branches then appear to continue without bound for

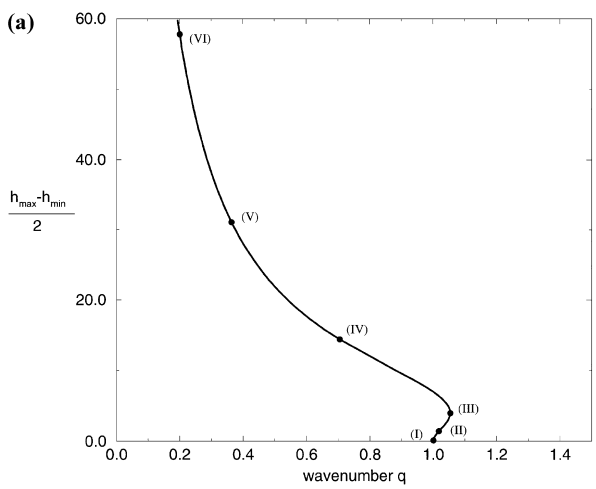

Fig. 6. Steady solution profiles for the segregation-free Riley–Davis equation. (a) Representative branch of solutions corresponding to $\kappa = 0, \delta = 0.3$. (b) Structure of solution profiles along the branch: sinusoidal structure near the bifurcation point (I, II); finger-type structure near the saddle node (III) qualitatively similar to the Sivashinsky equation; large-amplitude solutions characterized by flattened peaks and deep, narrow roots (IV–VI).

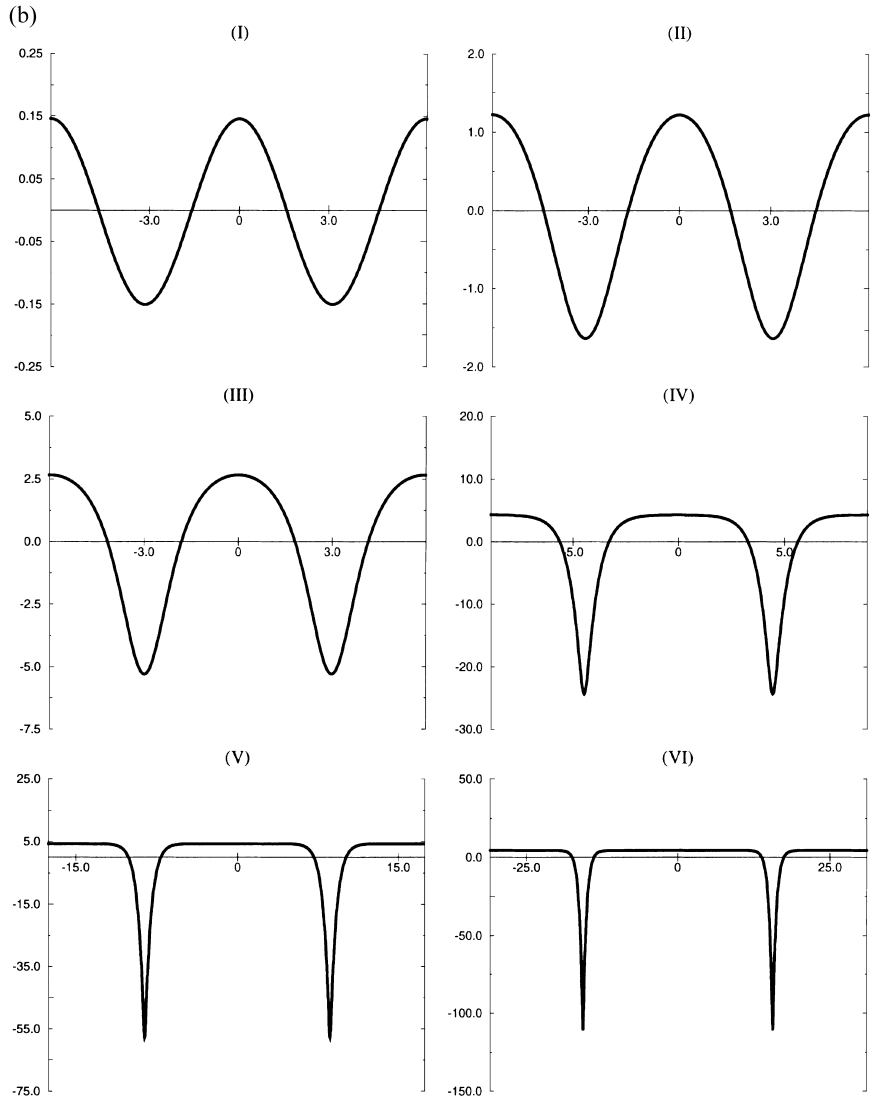

Fig. 6. (Continued)

increasing amplitude and wavelength. The numerical results indicate that as $\delta \to 0$, the amplitude at which the saddle node occurs scales as $1/\delta$, a result which is confirmed by balancing the first and last terms in (4.7).

The structure of these periodic solutions as one moves up the bifurcation curve is seen in Fig. 6, which shows a representative family of solution profiles for $\delta = 0.3$. Solutions close to the primary bifurcation point have the sinusoidal shape predicted by weakly nonlinear theory; as we continue up the branch, larger amplitude solutions begin to exhibit a finger-type structure qualitatively similar to the Sivashinsky equation. Beyond the saddle node, however, the effect of the additional nonlinearity is strongly felt, and we see that large-amplitude solutions are characterized by deep, narrow roots connected to long, nearly constant regions at the top of the profile. As wavelengths grow large, the width of the roots become asymptotically small compared to the period. In the remainder of the section, we shall seek an analytical understanding of this boundary-layer structure.

It is first convenient to rescale as follows:

$$\hat{x} = \frac{x}{\sqrt{\delta}}, \qquad \hat{h} = \delta h, \tag{4.9}$$

for which (4.7) becomes

$$\hat{h}_{\hat{x}\hat{x}\hat{x}\hat{x}} + \delta \hat{h}_{\hat{x}\hat{x}} = \left(\frac{\hat{h}^2}{2}\right)_{\hat{x}\hat{x}} - \left(\frac{\hat{h}_{\hat{x}}^2}{2}\right)_{\hat{x}\hat{x}}. \tag{4.10}$$

This equation may be integrated twice to yield

$$\hat{h}_{\hat{x}\hat{x}} + \left(\frac{\hat{h}_{\hat{x}}^2}{2}\right) = \frac{\hat{h}^2}{2} - \delta\hat{h} - c_1, \tag{4.11}$$

where c_1 is an arbitrary constant and we have exploited the periodicity to eliminate the second constant. This equation is of Riccati type; multiplying by $\mathrm{e}^{\hat{h}/2}$, it can be rewritten as

$$2(\mathrm{e}^{\hat{h}/2})_{\hat{x}\hat{x}} = \left(\frac{\hat{h}^2}{2} - \delta\hat{h} - c_1\right)\mathrm{e}^{\hat{h}/2}, \tag{4.12}$$

which has an integrating factor of the form $(\mathrm{e}^{\hat{h}/2})_{\hat{x}}$. Multiplying by this factor, integrating and changing variables eventually yields

$$(u_{\hat{x}})^2 + g(u) = b, \tag{4.13}$$

where

$$g(u) \equiv -u^2 + a\mathrm{e}^{-u}, \quad u \equiv \hat{h} - \delta - 1, \tag{4.14}$$

and a and b are constants of integration. We thus see that, as with the Sivashinsky equation, the Riley–Davis equation in the $\kappa = 0$ limit reduces to a potential problem, where in this case the potential $g(u)$ is the transcendental function given by (4.14). Unlike the Sivashinsky equation, the solution to (4.13), (4.14) may not be explicitly written in terms of elliptic integrals or known functions. However, we shall see that a leading-order description of the large-amplitude, deep-root solutions may be obtained using the method of asymptotic matching.

We observe that periodic solutions to (4.13), (4.14) exist only for $0 < a < 2/\mathrm{e}$, for which $g(u)$ has a potential well (see Fig. 7). Solutions oscillate between the two values of u obtained by intersecting the potential well with the line $y = b$. For each choice of a, there is exactly one solution within the potential well that satisfies the zero-mean requirement in h, and this determines b. As seen in Fig. 7, large-amplitude solutions exist for $a = \varepsilon \ll 1$. We thus consider

$$(u_{\hat{x}})^2 + g(u) = b, \quad g(u) \equiv -u^2 + \varepsilon\mathrm{e}^{-u}, \quad \varepsilon \ll 1, \tag{4.15}$$

with the requirement

$$\bar{u} \equiv \frac{1}{2L}\int_0^{2L} u \,\mathrm{d}\hat{x} = -(\delta + 1), \tag{4.16}$$

where $2L$ is the unknown interval length.

We seek a leading-order description of the solutions to (4.15) and (4.16) through the method of matched asymptotics, where the "outer" region includes the long, flat region at the top of the profile, and the "inner" region describes

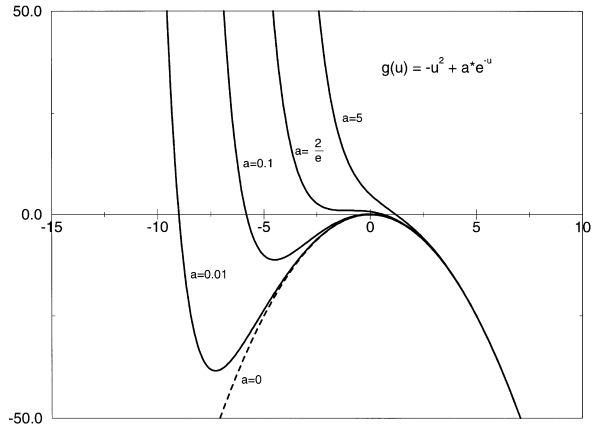

Fig. 7. Structure of the potential $g(u)$ for the steady, segregation-free Riley–Davis equation. Large-amplitude solutions exist for $a \ll 1$.

the corner layer in the deep root. We first determine the maximum and minimum values of the solution, which we shall refer to as u_+ and u_-, by computing the roots of

$$-u^2 + \varepsilon \mathrm{e}^{-u} = b. \tag{4.17}$$

Noting from (4.15) that

$$b \le g(u^*), \tag{4.18}$$

where u^* is the local maximum of $g(u)$. By differentiating and expanding the exponential for small u, we find

$$u^* = -\frac{\varepsilon}{2} - \frac{\varepsilon^2}{4} + \mathrm{O}(\varepsilon^3), \tag{4.19}$$

and

$$g(u^*) = \varepsilon + \frac{\varepsilon^2}{4} + \mathrm{O}(\varepsilon^3). \tag{4.20}$$

Observing from Fig. 7 the behavior of $g(u)$ as $\varepsilon \to 0$, it is reasonable to believe that b becomes arbitrarily close to $g(u^*)$ in this limit in order to satisfy the mean condition (4.16). By writing $b = \varepsilon + \varepsilon^2 \bar{b}$, where $\bar{b} \le 1/4$, we may now compute u_+ to leading order as a function of $\bar{b}$ and ε:

$$u_+ \sim -\varepsilon \left[\frac{1 + (1 - 4\bar{b})^{1/2}}{2} \right]. \tag{4.21}$$

The minimum value of the solution, u_-, is most easily found by writing

$$\varepsilon \mathrm{e}^{-u_-} = u_-^2 + b, \tag{4.22}$$

taking the natural logarithm of both sides, and solving by iteration. We find

$$u_- = -\ln\frac{1}{\varepsilon} - 2\ln\left(\ln\frac{1}{\varepsilon}\right) + \phi(\varepsilon), \tag{4.23}$$

where $\phi(\varepsilon) \to 0$ as $\varepsilon \to 0$.

The "outer" problem, which describes the profile in a neighborhood of u_+, is written by expanding the exponential term for small $|u|$ and neglecting terms of $\mathrm{O}(u^2)$ or higher:

$$u_{\hat{x}}^2 - u^2 + \varepsilon(1 - u) = b. \tag{4.24}$$

Thus

$$u_{\text{out}} \sim -\frac{\varepsilon}{2} - \sqrt{c}\cosh\hat{x}, \tag{4.25}$$

where $c \equiv \varepsilon + \varepsilon^2/4 - b$ and we have defined the maximum of the profile to be located at $\hat{x} = 0$.

To describe the "inner" region, we note that as the solution profile moves into the deep root, u grows in magnitude, and the exponential term in $g(u)$ becomes co-dominant. It is instructive to consider the more general problem

$$(u_{\hat{x}})^2 = -\varepsilon \mathrm{e}^{-u} + G(u), \tag{4.26}$$

where $G(u)$ is an arbitrary differentiable function of u, and u_- is defined by $G(u_-) = \varepsilon \mathrm{e}^{-u_-}$. We next let

$$u = u_- + \Delta, \tag{4.27}$$

which yields

$$(\Delta_x)^2 = u_-^2(1 - \mathrm{e}^{-\Delta}) + \mathrm{O}[|G'(u_-)|\Delta]. \tag{4.28}$$

It is now convenient to introduce the transformations

$$\Delta = -\ln v, \qquad \xi = |u_-|(L - \hat{x}). \tag{4.29}$$

This yields

$$(v_\xi)^2 = v^2 - v^3 + \mathrm{O}\left[\frac{|G'(u_-)||\ln v|v^2}{(u_-)^2}\right]. \tag{4.30}$$

Neglecting the error term, the solution is

$$v = \operatorname{sech}^2\left(\frac{\xi}{2}\right), \tag{4.31}$$

which corresponds to

$$u_{\text{in}} \sim u_- + 2\ln\left[\cosh\left(\frac{|u_-|(L - \hat{x})}{2}\right)\right]. \tag{4.32}$$

For our particular case, $G(u) \equiv 2u + b$; substituting this and (4.31) into (4.30), we may compute an expression for the error term:

$$\frac{|G'(u_-)||\ln v|\, v^2}{(u_-)^2} = \frac{4\operatorname{sech}^4(\xi/2)\ln[\cosh(\xi/2)]}{|u_-|}. \tag{4.33}$$

Note that since $\hat{x} = L$ corresponds to $\xi = 0$, the error term equals zero at $\hat{x} = L$.

Eq. (4.32) describes the profile in the deep root. Note that it is symmetric about $\hat{x} = L$ and (since $|u_-|$ is large) quickly approaches a sharply sloping linear profile as $\hat{x}$ moves away from L. This second observation may be easily seen by writing the hyperbolic function in terms of exponentials and neglecting the exponential which decays.

A leading-order match of the inner and outer solutions is now sought. We expand the outer solution (4.25) for $\hat{x} \gg 1,\ L - \hat{x} \ll 1$:

$$u_{\text{out}} \sim \frac{-\sqrt{c}\,\mathrm{e}^{\hat{x}}}{2} = \frac{-\sqrt{c}\,\mathrm{e}^{L}\,\mathrm{e}^{(\hat{x}-L)}}{2} \sim \frac{-\sqrt{c}\,\mathrm{e}^{L}}{2}[1 - (L - \hat{x}) + \cdots]. \tag{4.34}$$

A similar expansion is performed on the inner solution (4.32) for $|u_-|(L - \hat{x}) \gg 1$:

$$u_{\text{in}} \sim u_- + 2\ln\left[\frac{\exp(|u_-|(L - \hat{x}/2)}{2}\right] = u_- - \ln 4 + \frac{|u_-|(L - \hat{x})}{2}. \tag{4.35}$$

Note that since the matching region corresponds to $\xi \gg 1$, the magnitude of the error term (4.33) for the inner solution in this region is approximately

$$\frac{32\,\xi}{|u_-|\mathrm{e}^{2\xi}}, \tag{4.36}$$

which remains small compared to the inner solution.

Matching (4.34) and (4.35) at leading order determines the interval length:

$$u_- - \ln 4 = \frac{-\sqrt{c}\,\mathrm{e}^L}{2} \tag{4.37}$$

or

$$L = \ln\left[\frac{2(u_- - \ln 4)}{-\sqrt{c}}\right]. \tag{4.38}$$

Note from (4.23) that L approaches infinity as $\varepsilon \to 0$ and from (4.34) that the value of the outer solution (4.25) at $\hat{x} = L$ is given by

$$u_{\mathrm{out}}(L) = u_- - \ln 4 \tag{4.39}$$

plus an exponentially small correction. The remaining unknown, c, is determined by the mean condition (4.16). Normally, this would require integrating both the outer and inner solutions over their appropriate ranges of validity within the interval. In this case the task is simplified by noting from (4.39) that the difference between the inner and outer solutions at L to leading order is $\ln 4$, and as the boundary layer thickness is presumed small, the contribution to the integral in (4.16) from the inner solution is small, and $\bar{u}$ may be computed to leading order by integrating the outer solution from 0 to $2L$:

$$\bar{u} \equiv \frac{1}{2L}\int_0^{2L}\left(-\frac{\varepsilon}{2} - \sqrt{c}\cosh\hat{x}\right)\mathrm{d}\hat{x} = -(1+\delta), \tag{4.40}$$

which yields

$$\sqrt{c} \sim \frac{(1+\delta)L}{\sinh L}. \tag{4.41}$$

A boundary layer thickness of $1/L$ may now be deduced by substituting (4.41) into (4.38) and writing $\sinh L \sim \mathrm{e}^L$ for $L \gg 1$; simplifying the expression reveals $L \sim |u_-|$. Comparing this with the expression for the inner solution (4.32) yields the desired result.

Now, we are able to express a relationship between amplitude and wave number for large-amplitude solutions. Substituting (4.41) into (4.25) and using (4.39), we find

$$u_+ - u_- \sim u_{\mathrm{out}}(0) - [u_{\mathrm{out}}(L) + \ln 4] \sim (1+\delta)L - \ln 4. \tag{4.42}$$

In terms of the original variables x and h, this yields

$$\delta(h_{\max} - h_{\min}) \sim \frac{(1+\delta)\pi}{\sqrt{\delta}q} - \ln 4, \tag{4.43}$$

where q is the wave number. A suitable rescaling of the axes in Fig. 5 according to (4.43) reveals that all the solution branches have a common asymptote for large amplitude and approach a universal form as $\delta \to 0$ (cf. Fig. 8).

4.3. Results for $0 < \kappa < 1/4$

In Fig. 9, steady, periodic solution branches to the full Riley–Davis equation (2.3) are traced numerically for various nonzero values of κ and δ. As with the Sivashinsky equation, there are now two primary bifurcation points

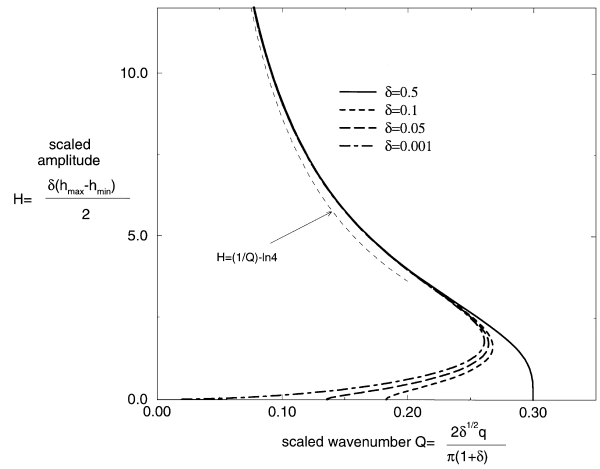

Fig. 8. Rescaled bifurcation diagram for the segregation-free Riley–Davis equation. As $\delta \to 0$, the branches approach a universal form as determined by Eq. (4.43).

q_+ and q_- for each $0 < \kappa < 1/4$, given by the roots of (4.1). Also analogous to the Sivashinsky equation, the branches which bifurcate from q_- appear to terminate at a point typically corresponding to a period n-tupling bifurcation on the q_+ branch, where in this case the value of n depends on both κ and δ. As the q_+ branches continue on to describe larger-amplitude solutions, it is these branches which are of primary interest here.

As seen in Fig. 9, the highly singular nature of the $\kappa \neq 0$ perturbation at larger amplitudes is clearly evident; amplitudes and wavelengths of steady solutions are now bounded. For κ sufficiently near 1/4 and δ sufficiently large, the q_+ and q_- branches may be directly connected, i.e. a branch which originates at q_+ terminates at q_-. This behavior is seen e.g. in Fig. 9(a) for $\kappa = 0.2, \delta = 1$. Of primary interest is the behavior for κ and δ sufficiently small. In either of these limits the maximum amplitude becomes large. After forming a saddle node corresponding to a maximum wavelength, the branches eventually end in a period-halving bifurcation. This is in sharp contrast to the $\kappa = 0$ case presented in the last section, in which amplitudes and wavelengths were unbounded, and also in sharp contrast to the Sivashinsky equation, in which the $\kappa \neq 0$ branches had the same leading-order behavior as the $\kappa = 0$ branch at large amplitude.

Solution profiles along a typical branch are presented in Fig. 10, corresponding to $\kappa = 0.1, \delta = 0.3$. As we move up the branch, the initially sinusoidal profiles begin to exhibit the deep-root, flattened-top features which are characteristic of the $\kappa = 0$ profiles. For sufficiently large amplitude, however, a dip forms at the top of the profile. As we continue along the branch, this dip grows in magnitude and each tip gradually separates into two distinct tips. Meanwhile, the magnitude of the deep root decreases. Eventually, the dip and the root are equal in height and the period-halving occurs. On the bifurcation diagram, if we draw a horizontal line through the terminal point and intersect this line with the increasing portion of the branch, the intersection point is located at precisely twice the wave number as the terminal point (points III and VIII of Fig. 10(a)). We find that these two points represent exactly the same solution. Thus, the period-halving bifurcation which occurs at the terminal point may also be considered a period-doubling bifurcation along the increasing portion of the branch.

A mathematical understanding of this behavior is aided by a closer study of the large-amplitude solutions for $\kappa = 0$. For that case, we see from (4.9),(4.14), and (4.21) that for fixed δ, as amplitudes grow asymptotically large ($\varepsilon \to 0$), the maximum h_+ of the solution profile approaches the order one value $1 + \delta^{-1}$, while the minimum h_- of the profile inside the deep root approaches $-\infty$. In order to satisfy the zero-mean requirement, the profile must remain near h_+ throughout most of the interval, resulting in the long, nearly constant region at the top of the profile as exhibited in Fig. 6(b) (VI). Since a deeper root can always be balanced by a correspondingly longer flat region, amplitudes and wavelengths can become unbounded. Note that when $\kappa = 0$, the Riley–Davis equation (2.3) admits

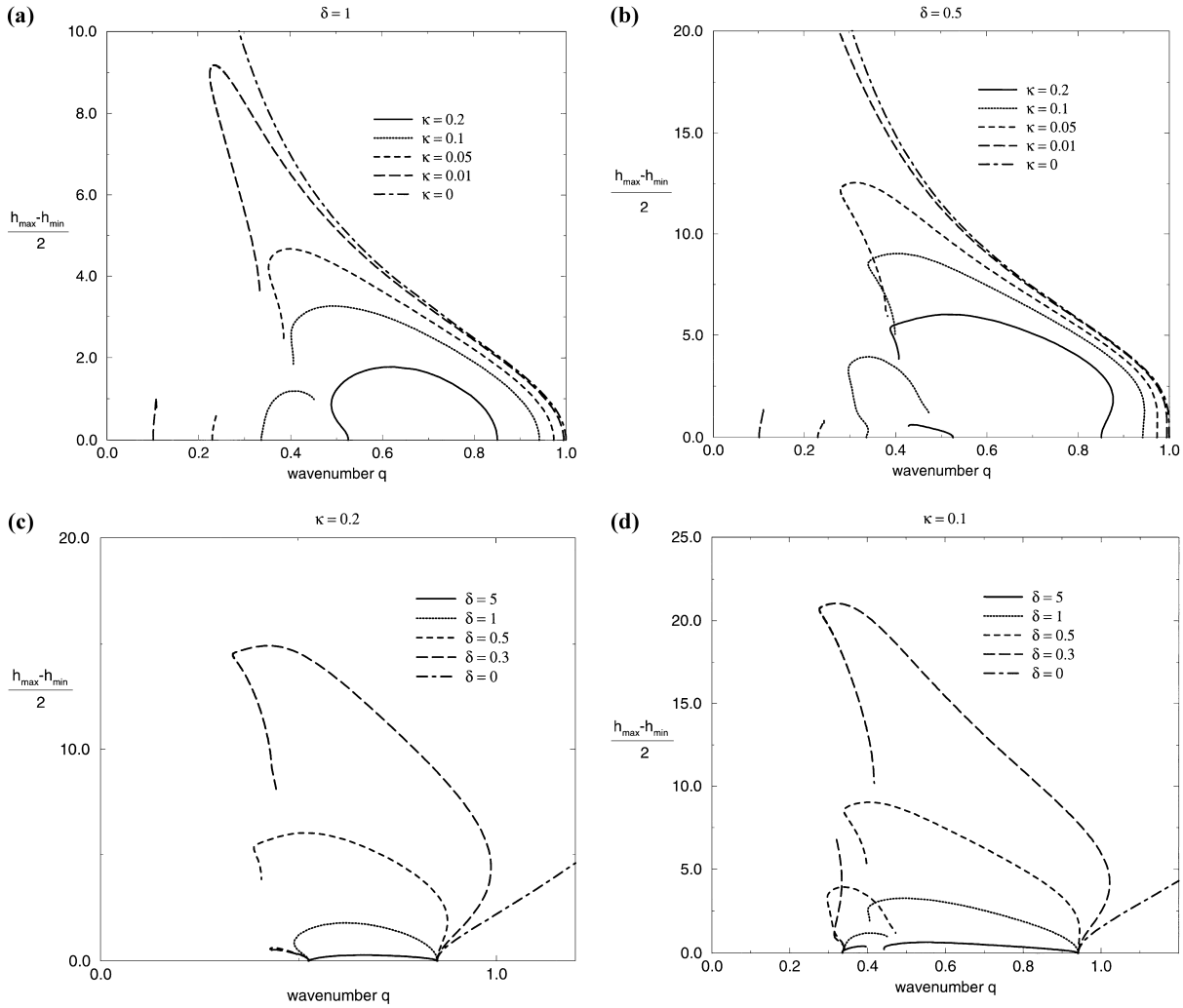

Fig. 9. Steady solution branches for the full Riley–Davis equation ($\kappa \neq 0$). Several representative branches are shown; the q_+ branches typically reach a maximum amplitude which becomes asymptotically large in both of the limits $\kappa \to 0$ and $\delta \to 0$; q_- branches typically terminate at smaller amplitudes or attach directly to the q_+ branches, as in $\kappa = 0.2$, $\delta = 1$.

any constant solution, and the flat region is allowed to persist. However, when $\kappa \neq 0$, no constant solution exists other than the trivial solution; any flat region which tries to form in order to maintain the zero-mean requirement is eventually offset by the κh term, which will always become co-dominant along any portion of the profile where derivatives are sufficiently small, and which does not allow a near-constant solution. Thus, the κh term causes a dip to form at the top of the profile. The zero-mean condition must be maintained by a corresponding decrease in the magnitude of the root, resulting in an upper bound in the amplitude of steady solutions.

Scaling laws are now sought for the maximum amplitude on the bifurcation curve as functions of small κ and δ. That is, we postulate

$$\max\left(\frac{h_{\max} - h_{\min}}{2}\right) \sim \kappa^r \tag{4.44}$$

as $\kappa \to 0$ with δ fixed, and

$$\max\left(\frac{h_{\max} - h_{\min}}{2}\right) \sim \delta^s \tag{4.45}$$

as $\delta \to 0$ with κ fixed, and numerically compute estimates for r and s. This analysis is made considerably difficult by numerical stiffness due to the disparate length scales in the large-amplitude, deep-root profiles. For example, with κ fixed at 0.227, we were able to trace the bifurcation curve beyond the maximum only for δ as small as 0.2; nevertheless, the scaling behavior did appear to be converging.

We first set $\delta = 1$ and traced bifurcation curves for several small values of κ, plotting $\max[(h_{\max} - h_{\min})/2]$ versus κ on a logarithmic scale (Fig. 11(a)), with $\kappa = 0.001$ corresponding to the smallest value for which the bifurcation curve could be traced beyond the maximum. We approximate r by measuring the slope of the line segment connecting the two data points corresponding to the two smallest values of κ in the figure, from which we conclude $r \approx -1.4$. Next, we approximated s by setting $\kappa = 0.227$, and considering several small values of δ (Fig. 11(b)). In a similar fashion, we conclude $s \approx -1.8$.

4.3.1. Linear stability analysis

In order to provide insight into the appearance and structure of deep-root interfaces in directional solidification experiments, as well as the phenomena of tip-splitting and tip-joining, the linear stability of several of the primary solution branches described in the previous section was analyzed numerically. The numerical method is described in detail in Appendix A; only the results are presented here. Four separate cases are examined, corresponding to supercritical with $\kappa = 0$, subcritical with $\kappa = 0$, supercritical with $\kappa \neq 0$, and subcritical with $\kappa \neq 0$.

Steady solutions are first computed numerically using a Newton method on the half interval $x \in [0, L]$, where we define $x = 0$ and $x = L$ to correspond to points along the profile where $h_x = 0$. Since a one-parameter family of solutions exist, we may compute a typical solution along the bifurcation curve by choosing a value for L and finding a solution which has the corresponding wavelength. Since the Riley–Davis equation is invariant under the transformations $x \to -x$ and $x \to x +$ constant, a full wavelength of the solution may be easily generated by reflecting the half-interval solution through either of the lines $x = 0$ or $x = L$. Because the primary branch undergoes a period-doubling bifurcation when $\kappa \neq 0$, we calculate the linear stability of solutions both to perturbations of

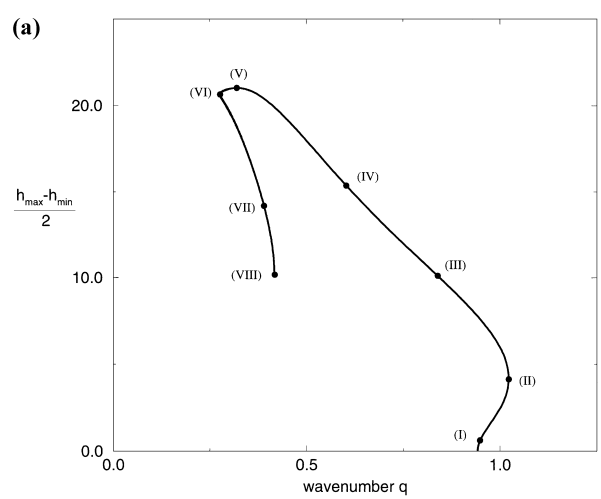

Fig. 10. Steady solution profiles for the full Riley–Davis equation ($\kappa \neq 0$). (a) Representative solution branch at q_+ corresponding to $\kappa = 0.1, \delta = 0.3$. (b) Structure of solution profiles along the branch: the maximum amplitude seen on the solution branch is related to a period-halving bifurcation via an indentation which forms at the top of the profile and deepens as one moves along the branch; this period-halving bifurcation may be related to tip-splitting and tip-joining phenomena which have been observed in the time-dependent problem.

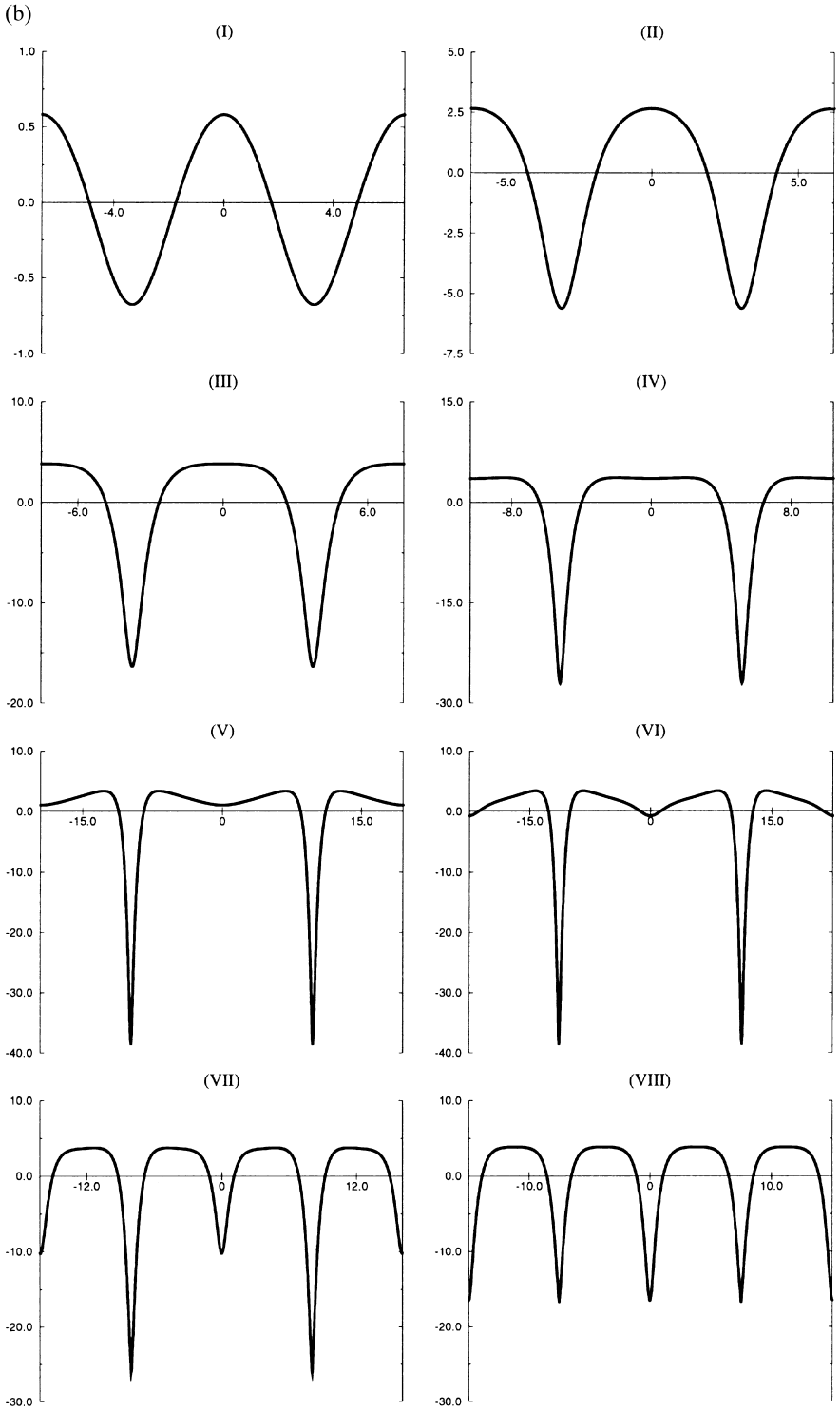

Fig. 10. (Continued)

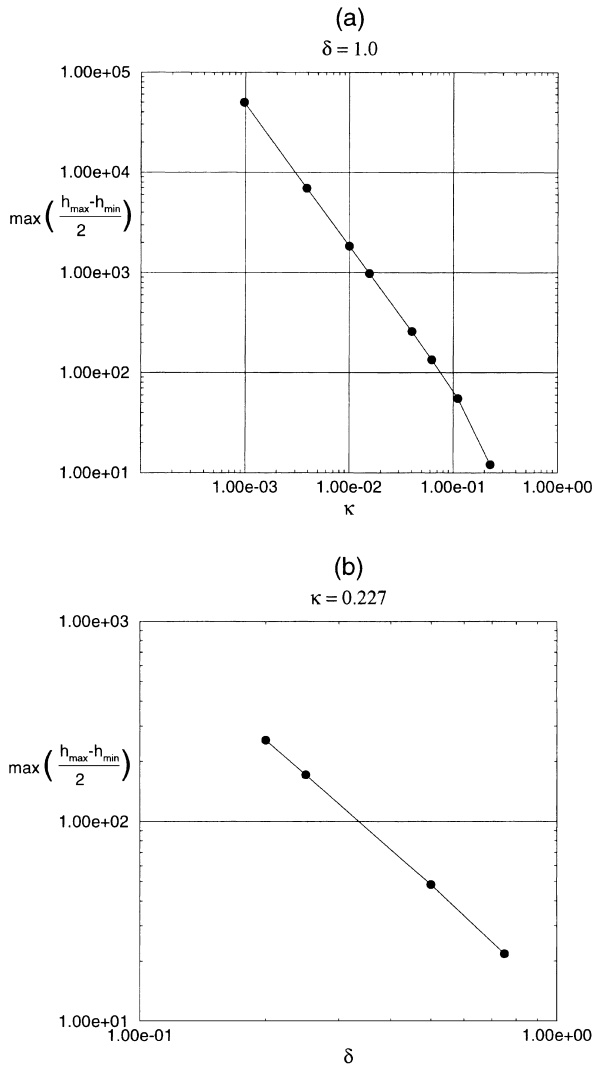

Fig. 11. Maximum amplitude of steady solutions. (a) δ fixed, $\kappa \to 0$; (b) κ fixed, $\delta \to 0$.

period $2L$ (*periodic modes*) and of period $4L$ (*period-doubling modes*). As the problem is solved on the half interval $[0, L]$, periodic modes can be expressed as linear combinations of those modes with even boundary conditions at both $x = 0$ and $x = L$ and those with odd boundary conditions at both points. Period-doubling modes can be expressed as linear combinations of the above modes plus modes with even–odd boundary conditions and odd–even boundary conditions. An instability of the steady solution exists if at least one mode has an eigenvalue with positive real part.

We first consider the segregation-free ($\kappa = 0$) branches. As a representative example of the supercritical regime, we consider the primary branch of solutions for $\delta = 1$. This branch is shown in Fig. 12(a).

The reader should recall that for $\kappa = 0$, the Riley–Davis equation has a two-parameter family of solutions, not all of which have zero mean. In investigating the stability of the zero-mean branch, we find that there is always an even–even mode with an eigenvalue of zero corresponding to a neutrally stable change in the mean of the solution. As we consider $\kappa = 0$ only as an asymptotic limit rather than a physically realizable situation, we do not consider

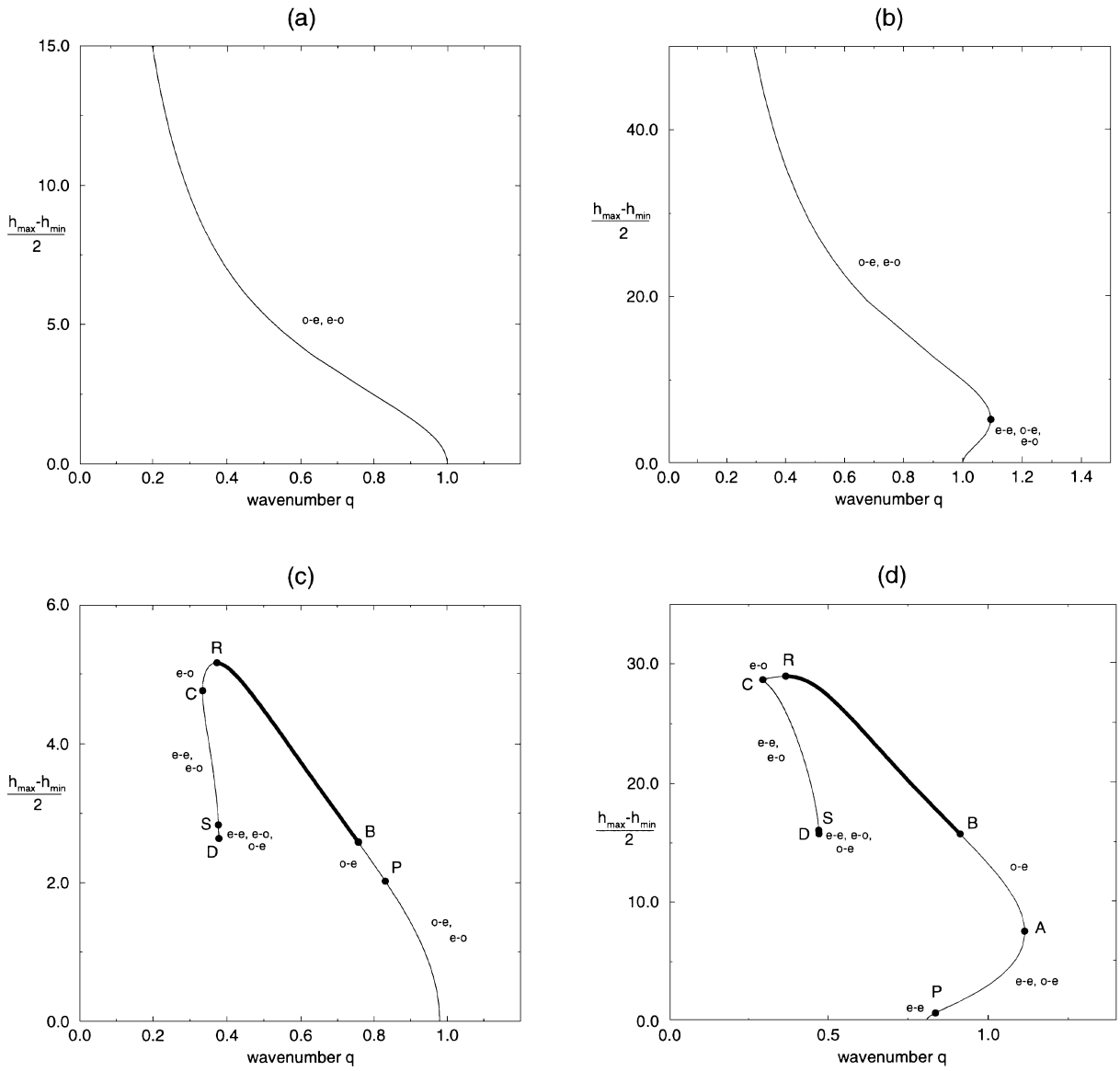

Fig. 12. Numerical linear stability results for the Riley–Davis equation. Perturbations are characterized by their boundary conditions at $x = 0$ and $x = L/2$: even–even (e–e) modes and odd–odd (o–o) modes, corresponding to perturbations of the same period as the steady solution, and odd–even (o–e) and even–odd (e–o) modes, corresponding to perturbations of twice the period. Four cases were tested: (a) $\kappa = 0$, supercritical; (b) $\kappa = 0$, subcritical; (c) $\kappa \neq 0$, supercritical; (d) $\kappa \neq 0$, subcritical. Each curve is labeled in various regions according to destabilizing modes which exist along that portion of the branch. The bold regions along the $\kappa \neq 0$ branch are stable to all periodic and period-doubling perturbations.

the nonzero mean solutions as playing any role in the solidification problem, and we attach no physical significance to the existence of this zero eigenvalue.

Also, we find that there is always an odd–odd mode with an eigenvalue of zero corresponding to a neutrally stable horizontal translation of the solution. No additional odd–odd modes were found to have positive eigenvalues for this or any of the cases that follow.

The even–even modes remain stable all along the branch for the supercritical case, except for the neutral mode mentioned above. Thus, the solutions on the primary branch are linearly stable to perturbations of the same period. However, we find there exists both an odd–even mode and an even–odd mode with a positive eigenvalue. The positive eigenvalue associated with the odd–even mode is larger than the one associated with the even–odd mode all along the branch, and both asymptote to zero but remain positive as we move high up the branch. Thus, the entire branch is linearly unstable to period-doubling perturbations.

We next consider the case $\kappa = 0, \delta = 1/4$, corresponding to the segregation-free, subcritical regime. The branch of zero-mean solutions is shown in Fig. 12(b), with the labeled point indicating the location of the saddle-node bifurcation.

Once again, there is both an even–even mode and an odd–odd mode with a zero eigenvalue, which have the same mathematical interpretation as in the supercritical case explained above. The primary difference in the subcritical case is that the branch is now initially unstable to an even–even mode. Also, there is a destabilizing odd–even mode and a destabilizing even–odd mode all along the branch. The odd–even mode is always the most destabilizing mode. At the saddle node, the branch gains a degree of stability as the eigenvalue associated with the even–even mode passes through zero. Beyond the saddle node, the branch is linearly stable to perturbations of the same period, but remains unstable to period-doubling perturbations; the eigenvalues associated with the odd–even and even–odd modes asymptote to zero but remain positive as in the supercritical case.

Thus the segregation-free branches have the property that any steady solution is unstable to a perturbation which tends to increase the period. In an experimental setting, one would hypothesize that the solution will coarsen to a wavelength determined by the domain size.

Next, we investigate the effect of segregation by considering solution branches for $\kappa \neq 0$. As a representative example of the supercritical regime, we first consider $\kappa = 0.04, \delta = 1$, shown in Fig. 12(c). In the figure, point C represents the saddle node which corresponds to the solution of maximum wavelength along the branch; point D is where the primary branch ends and period-halving occurs; point B is located by drawing a horizontal line through D and finding its intersection with the increasing portion of the branch. As discussed in the previous section, point B and point D represent exactly the same solution, and the period-halving bifurcation which occurs at point D may also be considered a period-doubling bifurcation at point B. Similar bifurcations are observed in Kuramoto–Sivashinsky [11] and other long-wave theories [12].

As with the $\kappa = 0$ branches, we find that there is always an odd–odd mode with a zero eigenvalue corresponding to a horizontal translation. However, since nonzero-mean steady solutions may not occur for $\kappa \neq 0$, there is no longer an even–even mode with a zero eigenvalue.

As expected, we find that the branch is initially stable to all even–even mode perturbations. However, there is initially both an odd–even mode and an even–odd mode which are destabilizing, with the odd–even mode having the larger positive eigenvalue. At point P, the branch gains a degree of stability as the even–odd eigenvalue passes through zero. At point B, the location of the period-doubling bifurcation, the odd–even eigenvalue passes through zero. Beyond this point, all of the eigenvalues remain negative until point R, located near the point of maximum amplitude. Thus, between point B and point R, the nontrivial steady solutions are linearly stable to all $2L$-periodic and $4L$-periodic perturbations.

Near the maximum (Point R), an even–odd eigenvalue passes through zero and becomes positive, so the branch is once again unstable to period-doubling bifurcations. At the saddle-node point C, a destabilizing even–even mode once again appears. These modes both remain destabilizing until the period-halving point D, with the even–odd mode being the more destabilizing of the two. Shortly before the period-halving point, a destabilizing odd–even mode appears at point S, so that the branch is now unstable to three modes.

Some intuition about the eigenvalues at the point D follows from the observation that the profile is merely two periods of the profile at the point B. The odd–even and even–odd eigenmodes are half-period translates of each other at this point, and consequently have the same eigenvalue. This eigenvalue is only slightly positive, and the eigenmode resembles the even–even mode (cf. Fig. 13(b)) except that it is localized on every other root. We associate the smallness of the eigenvalue with the localization on the roots.

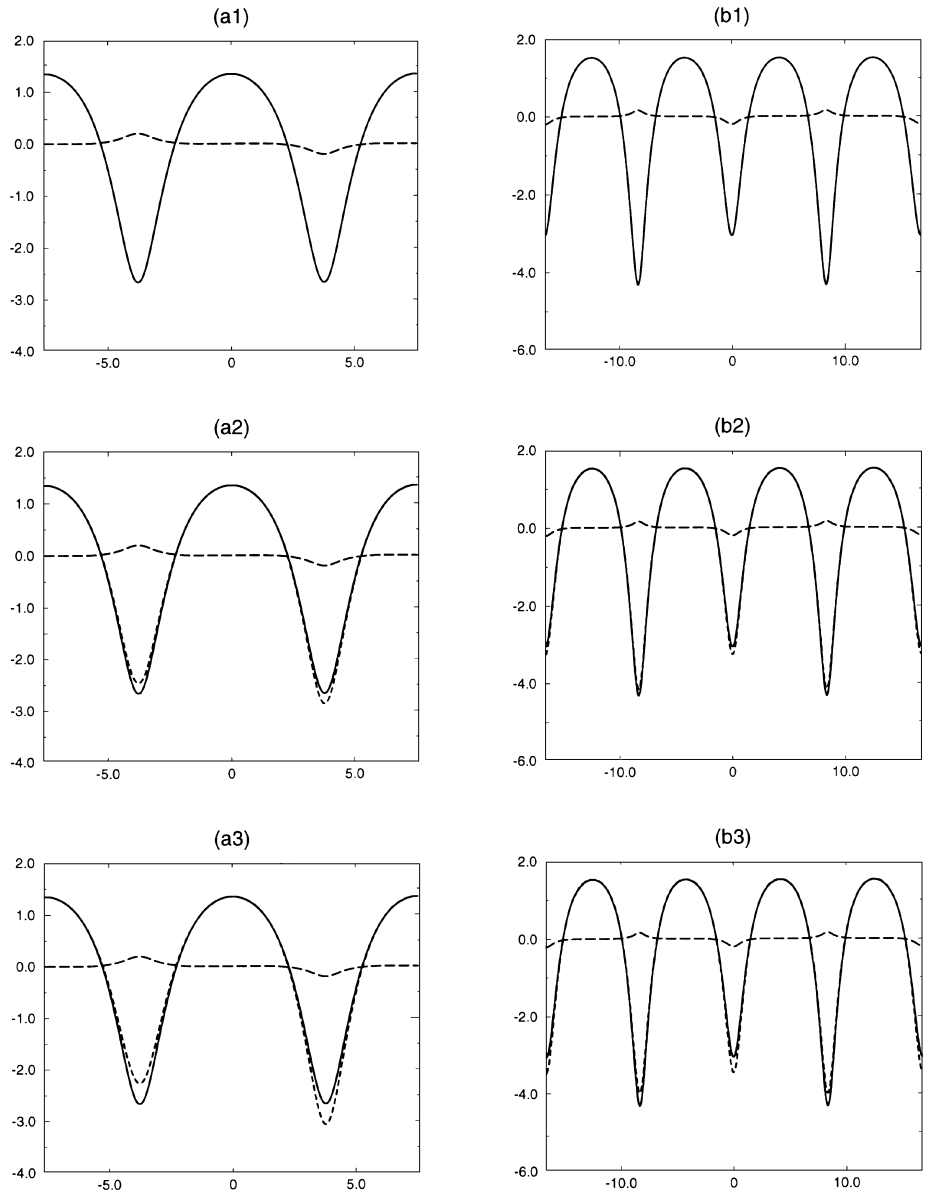

Fig. 13. Small-time effect of destabilizing modes; (a1)–(a3) a destabilizing odd–even mode (dashed line in (a1)) which may lead to tip-joining; (b1)–(b3) a destabilizing even–even mode which may lead to period-halving.

Finally, we consider the branch of solutions for $\kappa = 0.227$, $\delta = 0.2$, corresponding to the subcritical regime (Fig. 12(d)). Once again, we expect this branch to be distinguished from the supercritical case by the appearance of a destabilizing even–even mode near the bifurcation point, and we find this indeed to be the case. The branch is initially stable to all odd–even mode and even–odd modes, but a destabilizing odd–even mode appears at point P. This point corresponds to a period-doubling bifurcation in which the solution here is identical to the solution at the terminal point of the q_- branch. The q_+ branch loses a degree of stability to the q_- branch at this point. Shortly after P, this odd–even mode becomes the most destabilizing mode as its eigenvalue becomes larger than that of the even–even mode.

At the first saddle node (point A), the eigenvalue associated with the destabilizing even–even mode passes through zero and becomes negative, but the branch remains unstable to an odd–even mode until we reach the period-doubling point B. At this point, the odd–even eigenvalue passes through zero, and the branch is now stable to both 2L-periodic and 4L-periodic perturbations. Like the supercritical case, the subcritical branch loses stability at the point R with the appearance of a destabilizing even–odd mode, and loses another degree of stability at the second saddle node (point C) to a destabilizing even–even mode. These two modes remain destabilizing until the period-halving point D is reached. Just before the period-halving point, a destabilizing odd–even mode appears at point S, so that the branch is now unstable to three modes. As the maximum amplitude of the branch increases (with e.g. decreasing δ), the point S becomes nearly coincident with the point D and extremely difficult to elucidate numerically. We associate this behavior with the increasing localization of the eigenfunction as the depth of the roots increase.

An interesting and difficult question here deals with the relationship between a destabilizing mode and the experimentally observed phenomena of tip-splitting and tip-joining [16]. On the basis of our linear stability analysis, we can only make speculations on the basis of small-time behavior. As an example, we re-consider the supercritical branch $\kappa = 0.04$, $\delta = 1$ (Fig. 12(c)). Fig. 13 shows the small-time effect of destabilizing modes observed for two steady solutions along the branch. In Fig. 13(a1), a steady solution (solid line) along the increasing portion of the branch at $q \approx 0.83$ (near the point P) is shown, along with the destabilizing odd–even eigenmode observed at that point. Note that the deep roots in the steady solution are all of equal magnitude. We see in this case that the effect of the eigenmode after a small-time increment (Fig. 13(a2) and (a3)) is to cause alternating roots to deepen, and alternating roots to become shallower. This may lead to eventual tip-joining at larger times, and a saturation of the solution to a stable point on the bifurcation curve between points B and R corresponding to a larger amplitude and twice the wave number. We expect that because of the equal original depth of the roots, this result is independent of the sign of the eigenmode.

In Fig. 13(b), a steady solution at $q \approx 0.38$, near the terminal, period-halving point D, is shown, along with the destabilizing even–even eigenmode found at that point. Once again (and consistent with most of our observations of the numerical simulations), the effect of the eigenmode is most strongly felt near the roots of the steady solution. We see that in this case alternating deep roots of the steady solution are of unequal depth, and that the sign of the eigenmode is important. If the eigenmode is as shown in the figure, the small-time effect of the eigenmode is to relax the variation between the alternating roots (Fig. 13(b2) and (b3)). Period-halving is favored, and the solution is likely to saturate to a point between B and R corresponding to approximately twice the wave number.

The full Riley–Davis equation for $\kappa \neq 0$ clearly contains a wavelength selection mechanism not found in the segregation-free case. As the long-time effect of a perturbation may not be predicted by linear stability analysis, a full numerical simulation of the nonlinear, time-dependent Riley–Davis equation is warranted as a topic for future study. Due to the disparate length scales observed in the deep-rooted solutions, such a study is complicated by the need for a numerical scheme which makes extensive use of dynamic grid refinement. It should be noted that although grid refinement was used in our linear stability analysis, considerable numerical stiffness was encountered.

5. Summary and conclusions

Two models of interface evolution in directional solidification are studied: the Sivashinsky equation and the Riley–Davis equation. Both model the process in a long-wave limit, and their respective regimes of validity overlap. In the limit $\delta \to 0$, the Sivashinsky equation is recovered from the Riley–Davis equation. Physically, this limit corresponds to near-unity morphological parameter M, small segregation coefficient k, and large surface energy Γ, with Γk small.

The Sivashinsky equation has large-amplitude steady solutions which may be described in terms of elliptic functions. The description is exact when segregation effects are ignored ($\kappa = 0$), and asymptotic for $\kappa > 0$. These solutions have deep roots, but they are not confined to narrow boundary layers. The Sivashinsky equation, however,

is not a suitable model for studying the appearance of large-amplitude interfaces in solidification because all of its steady periodic solutions are unstable, and the equation typically exhibits finite-time blow-up.

The Riley–Davis limit may regularize this ill-posedness. The regularization lies in the incorporation of large surface energy into the model, which has a stabilizing effect on the solid–liquid interface. When $\delta \to 0$, the periodic solutions approach those of the Sivashinsky equation for fixed amplitudes; however, a new family of stable, periodic solutions appears when amplitudes scale like $1/\delta$ or larger. An unstable branch turns around in a saddle-node bifurcation, giving way to a new class of solutions characterized by deep roots confined to narrow boundary layers.

The existence and stability of these solutions is analyzed in detail in the limit of this regularizing process. For $\kappa = 0$ (and any δ), there exist steady solutions of arbitrarily small wave number and large amplitude. These solutions may be described asymptotically using boundary-layer theory, for which amplitudes are found to scale as the inverse of the wave number. Numerical linear stability analysis reveals that such solutions are always unstable to a perturbation of twice the wave number, i.e. a *period-doubling* perturbation. This instability suggests that coarsening will occur: on a bounded domain, adjacent tips will join until a single tip occupies the entire interval.

When $\kappa > 0$ the flattened tops become unstable, leading to a maximum possible amplitude. This introduces a small wave number cut-off and a range of stable periodic solutions. A preferred length scale is now contained in the system, and our linear stability results suggest that this length scale will be approached via a sequence of tip splitting or tip joining. In either of the limits $\kappa \to 0$ or $\delta \to 0$ the stable range approaches arbitrarily large amplitude/small wave number. This suggests that far from marginal stability, deep, finger-type solutions are preferred, in agreement with experiment [28].

While the Riley–Davis equation is a long-wave model whose asymptotic assumptions are violated by the large gradients exhibited in these deep-root solutions, this analysis provides an ideal workbench for studying the appearance and dynamics of finger-type interfaces in directional solidification.

Future research on this problem is warranted. Our numerical linear stability analysis was limited to the study of periodic and period-doubling perturbations. An investigation of the complete spectrum of superharmonic instabilities, akin to [11,15], is suggested. A full time-dependent numerical simulation will provide further insight into the wavelength selection process, but a need for dynamical grid refinement to overcome numerical stiffness is suggested by the disparate length scales in the deep-root solutions. In addition, the open problems of quantifying coarsening (in the absence of segregation effects) and proving global existence and deriving uniform bounds on the temporal evolution (in the presence of segregation effects) should also provide fertile grounds for future analysis.

Acknowledgements

The authors would like to thank Andrea Bertozzi, Steve Davis, Geoff McFadden, Philippe Metzener, Mary Silber and an anonymous referee for insightful comments about this research. AJB would also like to thank Edgar Knobloch for his sometimes overlooked keen insight and patient commentary on this and other research. This research has been supported by NSF grant DMS-9306326. Lou Rossi was supported by an NSF Mathematical Sciences Postdoctoral Research Fellowship (DMS-9407660). AJB would like to acknowledge the generous support and hospitality of MSRI.

Appendix A. Numerical techniques

We numerically approximated the Riley–Davis and Sivashinsky equations (2.3) and (2.4) to determine the asymptotic scaling and linear stability of steady solutions discussed in Section 3 and 4. The steady solutions are approximated with second order finite differences in conservative form. Since we are only interested in $2L$-periodic solutions that are symmetric about $x = L$, we computed a solution on $[0, L]$ with even boundary conditions. A mesh refinement scheme is used to boost the global accuracy of the approximation when h and its derivatives are large. The eigenmodes of steady solutions are computed using both Hessenberg decomposition and shifted inverse iteration techniques.

A.1. Steady solutions

In order to approximate the steady Riley–Davis equation (2.3), the original PDE is discretized into a finite linear system:

$$-\Delta^4 h_i - \Delta^2 h_i - \kappa h_i + \Delta^2 \left[\frac{h_i^2}{2} - \delta \frac{(\Delta_0 h_i)^2}{2} \right] = 0, \tag{A.1}$$

where $h_i = h[i(\mathrm{d}x)]$, Δ^2 signifies a second finite difference,

$$\Delta^2 f_i = \frac{f_{i+1} - 2f_i + f_{i-1}}{(\mathrm{d}x)^2},$$

$(\mathrm{d}x) = L(N-1)$, and Δ_0 signifies a centered difference

$$\Delta_0 f_i = \frac{f_{i+1} - f_{i-1}}{2(\mathrm{d}x)}.$$

This finite difference equation is in conservative form to assure that truncation errors do not alter the natural zero-mean quality of $\kappa \neq 0$ solutions. Since this equation is translationally invariant, even boundary conditions at each endpoint constrain the solution so that the extrema associated with nontrivial solutions close to the bifurcation point lie at either endpoint. We solve the nonlinear discrete system (A.1) using Newton–Raphson iterations with the l^2 norm to measure the quality of the solution. The resulting banded pentadiagonal system is solved using LAPACK subroutines [1].

Since large-amplitude solutions have deep, narrow roots, we implemented a grid refinement process in the root to boost the accuracy of the solution in those areas where h and derivatives of h are large. Global refinement is not an efficient procedure because increasing N globally only leads to changes in h in the narrow region of the trough. To solve this problem, a "top level" solution, $h^{(0)}$, is computed as described above on the whole domain, $[0, L]$. Assuming this narrow root is on the right side of the domain, we determine the "next level" solution, $h^{(1)}$, on the domain $[L/2, L]$ using N mesh points. This requires the same computational effort as the top level, but achieves twice the resolution on half of the domain. Then, the nonlinear system (A.1) is solved on this newly defined domain using Newton–Raphson iterations with even boundary conditions on the right and Dirichlet boundary conditions on the left (at $x = L/2$), derived from $h^{(0)}$ using fourth order interpolation onto the finer mesh. The process can be repeated as many times as one desires to accurately determine the depth of the trough. For the more extreme parameter values discussed in this paper, three levels of refinement were sufficient for values of $N = 800$.

When $\kappa = 0$, an extra parameter, the integral of h, is introduced into the discrete problem similar to the continuous PDE as discussed earlier in the paper. In this case, Eq. (A.1) is integrated twice and solved for arbitrary C,

$$\Delta^2 h_i - h_i + \left[\frac{h_i^2}{2} - \delta \frac{(\Delta_0 h_i)^2}{2} \right] = C. \tag{A.2}$$

We determine the value of C corresponding to a zero-mean solution by a secant iteration method. Thus, we determine steady solutions when $\kappa = 0$ in a doubly iterative process. The inner iteration involves Newton–Raphson iterations to solve the nonlinear system (A.2). The outer iteration varies C to find a value which results in a zero-mean solution.

A.2. Linear stability

To approximate growth rates of different disturbances to the unsteady Riley–Davis equations, we find solutions of the form

$$h_i(t) = \bar{h}_i + h_i' \mathrm{e}^{\lambda t}, \tag{A.3}$$

where $\bar{h}$ is a solution to the steady Riley–Davis equations. For the approximation, $\bar{h}_i$ is a solution of Eq. (A.1) or (A.2) for $\kappa \neq 0$ or $\kappa = 0$, respectively. A semidiscrete approximation to the full Riley–Davis equation is

$$(\partial_t - \delta\partial_t\Delta^2)h_i = -\Delta^4 h_i - \Delta^2 h_i - \kappa h_i + \Delta^2\left[\frac{h_i^2}{2} - \delta\frac{(\Delta_0 h_i)^2}{2}\right], \tag{A.4}$$

where h_i is now a function of time. Substituting Eq. (A.3) into (A.4), we obtain the following discrete linear system,

$$\lambda(I - \delta\Delta^2)h_i' = -\Delta^4 h_i' - \Delta^2 h_i' - \kappa h_i' + \Delta^2[\bar{h}_i h_i' - \delta(\Delta_0\bar{h}_i\Delta_0 h_i')]. \tag{A.5}$$

We solve this system using both the LAPACK eigensystem solvers which determine the entire spectrum, and a shifted inverse iterative solver to track the eigenvalue with the largest real part ($\Re(\lambda) > \Re(\lambda_i)$ for all i) and its corresponding eigenfunction [1].

While the steady solutions, $\bar{h}$, are $2L$-periodic functions symmetric about $x = L$, $\bar{h}$ may be unstable to arbitrary disturbances. In this paper, we focus only on $2L$-periodic and $4L$-periodic (period doubling) disturbances. In these cases, it is possible to study the stability of $\bar{h}$ to arbitrary $2L$-periodic and $4L$-periodic disturbances by solving (A.5) on $[0, L]$ by manipulating the boundary conditions at the left and right endpoints.

All $2L$-periodic disturbances to $\bar{h}$ are linear combinations of solutions to (A.5) with either even left together with even right (even–even) boundary conditions or odd left together with odd right (odd–odd) boundary conditions. These restrictions arise from the symmetries of $\bar{h}$ about $x = 0$ and $x = L$ as well as the symmetries of Eq. (A.5). Assuming $\bar{h}$ is a solution as described in the previous subsection and $\tilde{h}(x)$ is any $2L$-periodic solution to (A.5), then $\tilde{h}(-x)$ is also a solution to Eq. (A.5). Rather than directly computing $\tilde{h}(x)$ on $[0, 2L]$, one can solve for $h_{\mathrm{ee}}(x)$ and $h_{\mathrm{oo}}(x)$ on $[0, L]$ where these functions have even–even and odd–odd boundary conditions, respectively. The symmetries noted above guarantee that these two functions, when extended onto $[0, 2L]$, span the space of all $2L$-periodic disturbances because

$$\tilde{h} = h_{\mathrm{ee}}(x) + h_{\mathrm{oo}}(x), \quad h_{\mathrm{ee}}(x) = \frac{\tilde{h}(x) + \tilde{h}(-x)}{2}, \quad h_{\mathrm{oo}}(x) = \frac{\tilde{h}(x) - \tilde{h}(-x)}{2}. \tag{A.6}$$

Similarly, all $4L$-periodic disturbances to $\bar{h}$ are linear combinations of solutions to (A.5) with even–odd, odd–even in addition to the even–even and odd–odd boundary conditions which are captured in the $2L$-periodic analysis. Since $\bar{h}$ is $2L$-periodic with symmetry about $x = L$, there are additional symmetries to $4L$-periodic solutions of (A.5). If $\tilde{h}$ is a solution, so is $\tilde{h}(2L + x)$ and $\tilde{h}(2L - x)$ in addition to $\tilde{h}(-x)$ as in the $2L$-periodic case. These symmetries guarantee that $h_{\mathrm{eo}}(x)$ and $h_{\mathrm{oe}}(x)$ on $[0, L]$, in addition to $h_{\mathrm{ee}}(x)$ and $h_{\mathrm{oo}}(x)$ from the $2L$-periodic case, will span the space of $4L$-periodic disturbances $\tilde{h}(x)$. The two new functions, h_{eo} and h_{oe}, have even–odd and odd–even boundary conditions on $[0, L]$, and when extended onto $[0, 4L]$, have the following $2L$-periodicity:

$$h_{\mathrm{eo}}(x + 2L) = -h_{\mathrm{eo}}(x), \qquad h_{\mathrm{oe}}(x + 2L) = -h_{\mathrm{oe}}(x).$$

Hence, any $4L$-periodic disturbance $\tilde{h}$ can be expressed as a linear combination of $h_{\mathrm{ee}}, h_{\mathrm{eo}}, h_{\mathrm{oe}}, h_{\mathrm{oo}}$ on an interval $[0, L]$, saving substantial amounts of computational effort.

References

[1] E. Anderson et al., LAPACK User's Guide, 2nd ed., SIAM, Philadelphia, PA, 1995.

[2] A.J. Bernoff, A. Bertozzi, Singularities in a modified Kuramoto–Sivashinsky equation describing interface motion for phase transition, Physica D 85 (1995) 375–404.

[3] K. Brattkus, S.H. Davis, Cellular growth near absolute stability, Phys. Rev. B 38 (1988) 452–460.

[4] P.F. Byrd, M.D. Friedman, Handbook of Elliptic Integrals for Engineers and Scientists, 2nd ed. (rev.), Springer, New York, NY, 1971.

[5] B. Caroli, C. Caroli, B. Roulet, On the emergence of one-dimensional front instabilities in directional solidification and fusion of binary mixtures, J. Phys. 43 (1982) 1767–1780.

[6] B. Caroli, C. Caroli, B. Roulet, The Mullins–Sekerka instability in directional solidification of thin samples, J. Crystal Growth 76 (1986) 31–49.
[7] S.R. Coriell, G.B. McFadden, Morphological stability, in: D.T.J. Hurle (Ed.), Handbook of Crystal Growth, Elsevier, Amsterdam, 1993, pp. 785–857.
[8] S.R. Coriell, R.F. Sekerka, Oscillatory morphological instabilities due to non-equilibrium segregation, J. Crystal Growth 61 (1983) 499–508.
[9] J.M. Hyman, A. Novick-Cohen, P. Rosenau, Modified asymptotic approach to modeling a dilute-binary-alloy solidification front, Phys. Rev. B 37 (1988) 7603–7608.
[10] M.D. Impey, D.S. Riley, A.A. Wheeler, Bifurcation analysis of cellular interfaces in unidirectional solidification of a dilute binary mixture, SIAM J. Appl. Math. 53 (1993) 78–95.
[11] I.G. Kevrekidis, B. Nicolaenko, J.C. Scovel, Back in the saddle again: a computer-assisted study of the Kuramoto–Sivashinsky equation, SIAM J. Appl. Math. 50 (1990) 760–790.
[12] E. Knobloch, Pattern selection in long-wavelength convection, Physica D 41 (1990) 450–479.
[13] J.S. Langer, Instabilities and pattern formation in crystal growth, Rev. Modern Phys. 52 (1980) 1–28.
[14] P. Metzener, Dynamics in directional solidification at nonequilibrium, in: Y.Y. Renardy et al. (Eds.), Advances in Multi-fluid Flows, SIAM, Philadelphia, PA, 1996, pp. 131–157.
[15] C. Misbah, H. Müller-Krumbhaar, D.E. Temkin, Interface structure at large supercooling, J. Phys. I 1 (1991) 585–601.
[16] L.R. Morris, W.C. Winegard, The development of cells during the solidification of a dilute Pb–Sb alloy, J. Crystal Growth 5 (1969) 361–375.
[17] W.W. Mullins, R.F. Sekerka, Stability of a planar interface during solidification of a dilute binary alloy, J. Appl. Phys. 35 (1964) 444–451.
[18] A. Nadarajah, R. Narayanan, Morphological instability in dilute binary systems – a uniform approach, PCH PhysicoChem. Hydrod. 11 (1989) 81–102.
[19] A. Novick-Cohen, Interfacial instabilities in directional solidification of dilute binary alloys: the Kuramoto–Sivashinsky equation, Physica D 26 (1987) 403–410.
[20] A. Novick-Cohen, Instability of periodic states for the Sivashinsky equation, Quart. Appl. Math. 48 (1990) 217–224.
[21] A. Novick-Cohen, On Cahn–Hilliard type equations, Nonlinear Anal. TMA 15 (1990) 797–814 .
[22] A. Novick-Cohen, Blow-up and growth in the directional solidification of dilute binary alloys, Appl. Anal. 47 (1992) 241–257.
[23] A. Novick-Cohen, M. Grinfeld, Counting the stationary states of the Sivashinsky equation, Nonlinear Anal. TMA 24 (1995) 875–881.
[24] A. Novick-Cohen, L.A. Peletier, The steady states of one-dimensional Sivashinsky equations, Appl. Math. Lett. 4 (1991) 69–71.
[25] A. Novick-Cohen, L.A. Peletier, The steady states of one-dimensional Sivashinsky equations, Quart. Appl. Math. 50 (1992) 759–777.
[26] A. Novick-Cohen, G.I. Sivashinsky, On the solidification front of a dilute binary alloy: thermal diffusivity effects and breathing solutions, Physica D 20 (1986) 237–258.
[27] B. Palais, Blowup for nonlinear equations using a comparison principle in Fourier space, Commun. Pure Appl. Math. XLI (1988) 165–196.
[28] P. Pelcé, Dynamics of curved fronts, Perspectives in Physics, Academic Press, New York, 1988.
[29] D.S. Riley, S.H. Davis, Long-wave morphological instabilities in the directional solidification of a dilute binary mixture, SIAM J. Appl. Math. 50 (1990) 420–436.
[30] D.C. Sarocka, A.J. Bernoff, An intrinsic equation of interfacial motion for the solidification of a pure hypercooled melt, Physica D 85 (1995) 348–374.
[31] G.I. Sivashinsky, On cellular instability in the solidification of a dilute binary alloy, Physica D 8 (1983) 243–248.
[32] A.C. Skeldon, G.B. McFadden, M.D. Impey, D.S. Riley, K.A. Cliffe, A.A. Wheeler, S.H. Davis, On long-wave morphological instabilities in directional solidification, Eur. J. Appl. Math. 6 (1995) 639–652.
[33] R. Sriranganathan, D.J. Wollkind, D.B. Oulton, A theoretical investigation of the development of interfacial cells during the solidification of a dilute binary alloy: Comparison with the experiments of Morris and Winegard, J. Crystal Growth 62 (1983) 265–283.
[34] L.H. Ungar, M.J. Bennett, R.A. Brown, Cellular interface morphologies in directional solidification III: the effects of heat transfer and solid diffusivity, Phys. Rev. B 31 (1985) 5923–5930.
[35] J. Vinals, R.F. Sekerka, P.P. Debroy, Morphological stability analysis of directional solidification in thin samples with lateral heat transfer, J. Crystal Growth 89 (1988) 405–414.
[36] D.J. Wollkind, L.A. Segel, A nonlinear stability analysis of the freezing of a dilute binary alloy, Phil. Trans. R. Soc. London A 268 (1970) 351–380.
[37] G.W. Young, S.H. Davis, Directional solidification with buoyancy in systems with small segregation coefficient, Phys. Rev B 34 (1986) 3388–3396.
[38] G.W. Young, S.H. Davis, Morphological instabilities in directional solidification of a binary alloy: end effects, SIAM J. Appl Math. 49 (1989) 152–164.
[39] G.W. Young, S.H. Davis, K. Brattkus, Anisotropic interface kinetics and tilted cells in unidirectional solidification, J. Crystal Growth 83 (1987) 560–571.